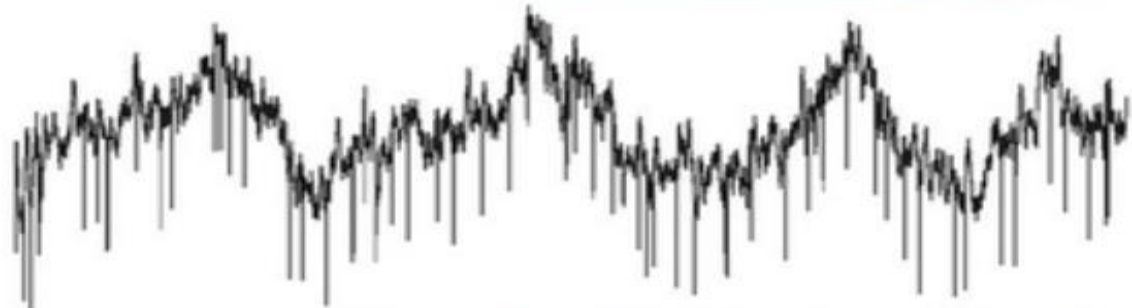


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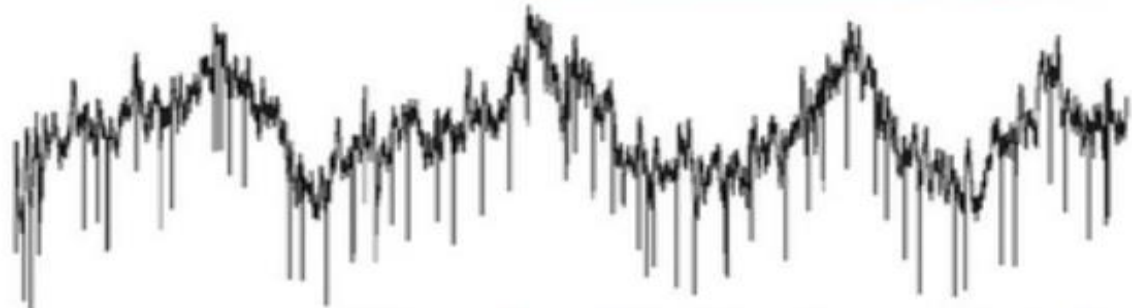
Electrical recordings of brain activity: Introduction to neural computation

Nicolas Catz
CRPN, AIX-MARSEILLE UNIVERSITÉ



Electrical recordings of brain activity: From Raw signal to spike trains

Nicolas Catz
CRPN, AIX-MARSEILLE UNIVERSITÉ



The origin of extracellular fields and currents — EEG, ECoG, LFP and spikes

György Buzsáki^{1,2,3}, Costas A. Anastassiou⁴ and Christof Koch^{4,5}

Abstract | Neuronal activity in the brain gives rise to transmembrane currents that can be measured in the extracellular medium. Although the major contributor of the extracellular signal is the synaptic transmembrane current, other sources — including Na^+ and Ca^{2+} spikes, ionic fluxes through voltage- and ligand-gated channels, and intrinsic membrane oscillations — can substantially shape the extracellular field. High-density recordings of field activity in animals and subdural grid recordings in humans, combined with recently developed data processing tools and computational modelling, can provide insight into the cooperative behaviour of neurons, their average synaptic input and their spiking output, and can increase our understanding of how these processes contribute to the extracellular signal.

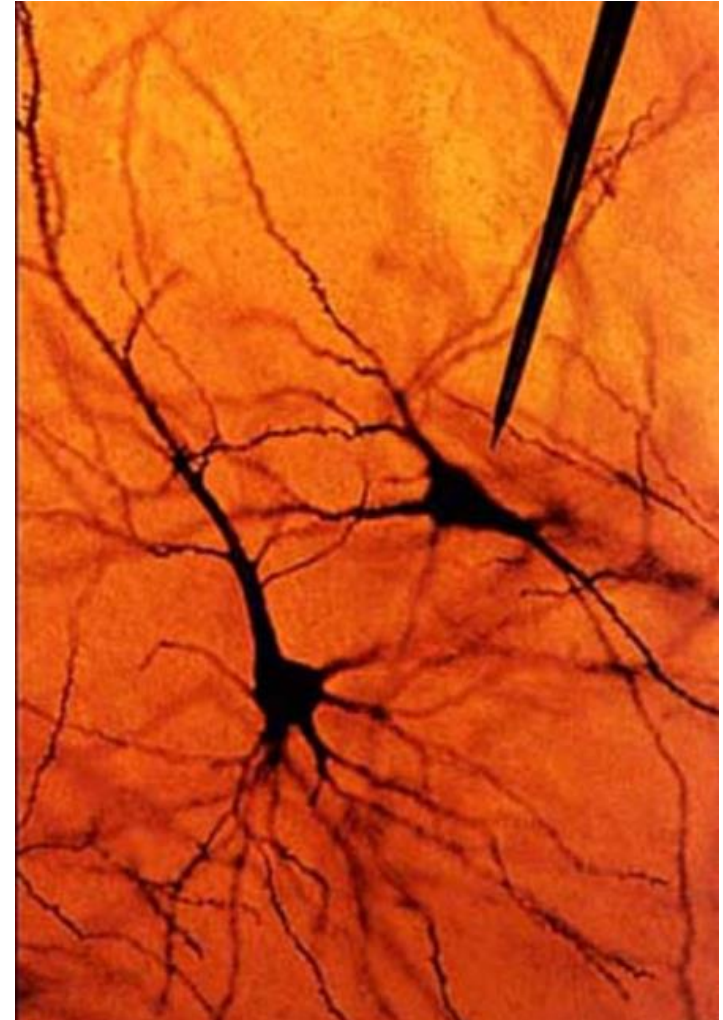
Extracting information from neuronal populations: information theory and decoding approaches

Rodrigo Quijano Quiroga^{} and Stefano Panzeri^{†§}*

Abstract | To a large extent, progress in neuroscience has been driven by the study of single-cell responses averaged over several repetitions of stimuli or behaviours. However, the brain typically makes decisions based on single events by evaluating the activity of large neuronal populations. Therefore, to further understand how the brain processes information, it is important to shift from a single-neuron, multiple-trial framework to multiple-neuron, single-trial methodologies. Two related approaches — decoding and information theory — can be used to extract single-trial information from the activity of neuronal populations. Such population analysis can give us more information about how neurons encode stimulus features than traditional single-cell studies.

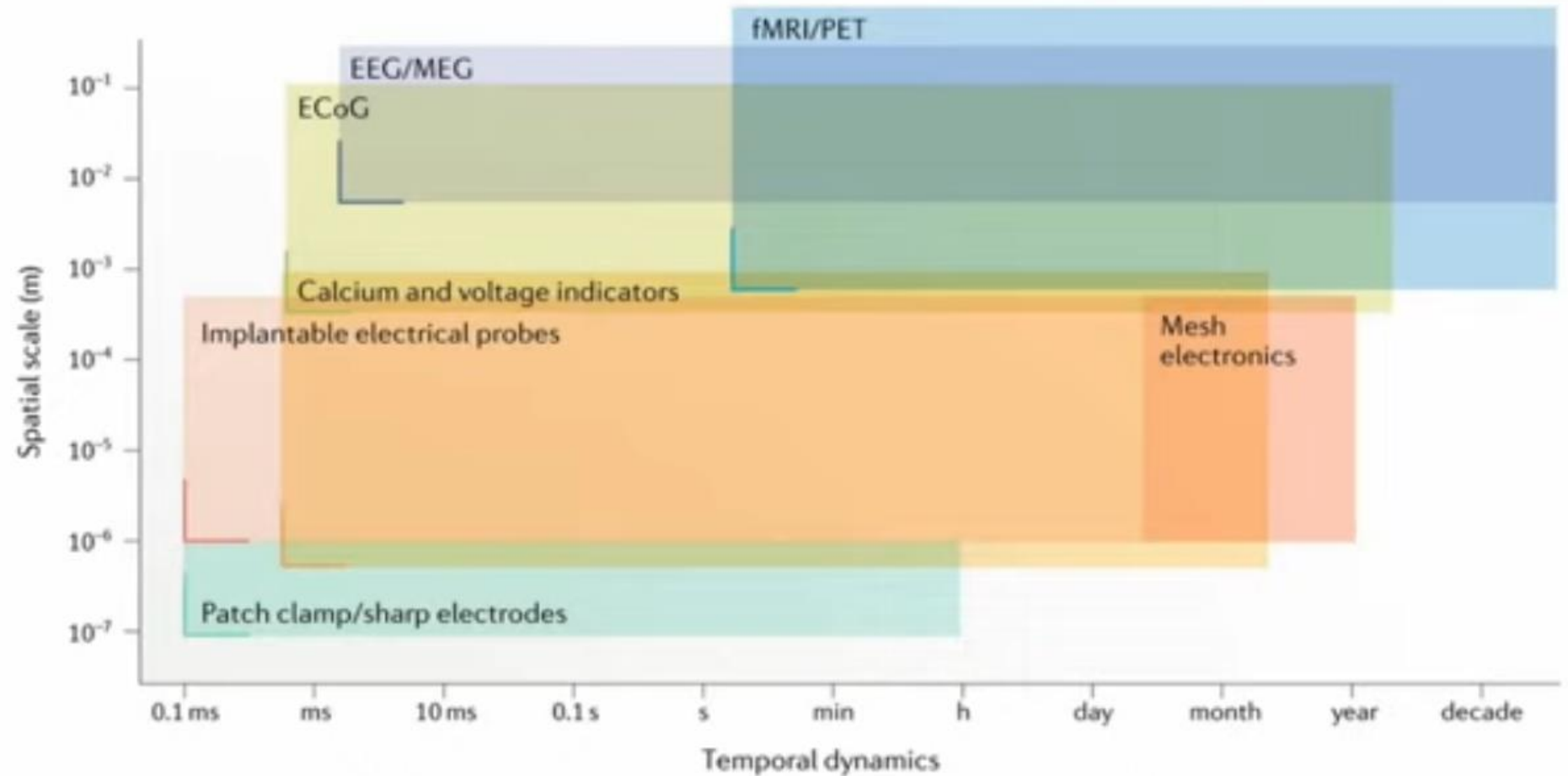
What is the origin of extracellular electrical signals ?

- Electrical recording in the brain are made with electrodes
- Recordings can be made inside of single cells
- Most often, in behaving animals, recordings are made of signals outside of neurons



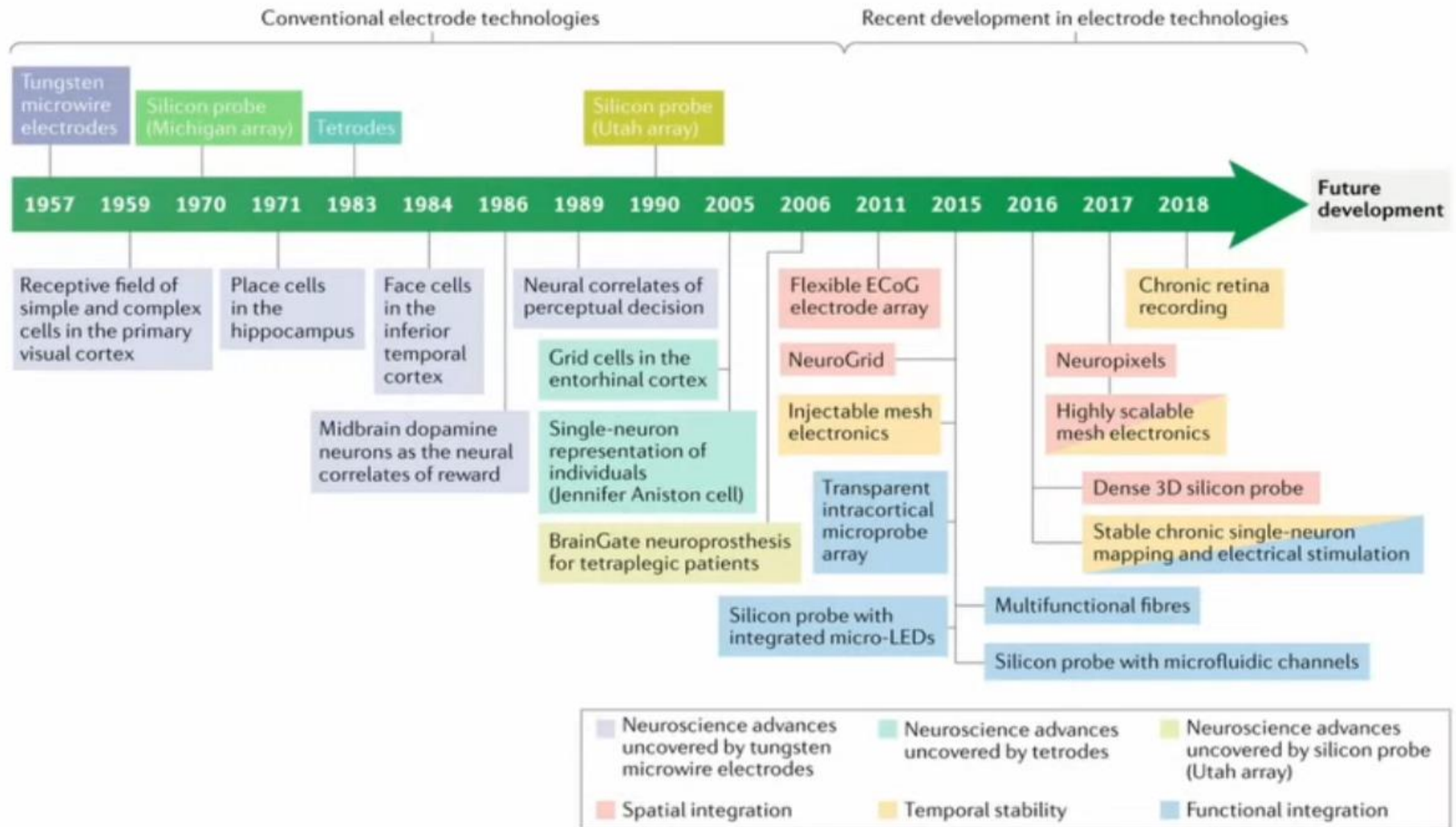
What is the origin of extracellular electrical signals ?

Spatiotemporal limitations of existing neurotechnologies



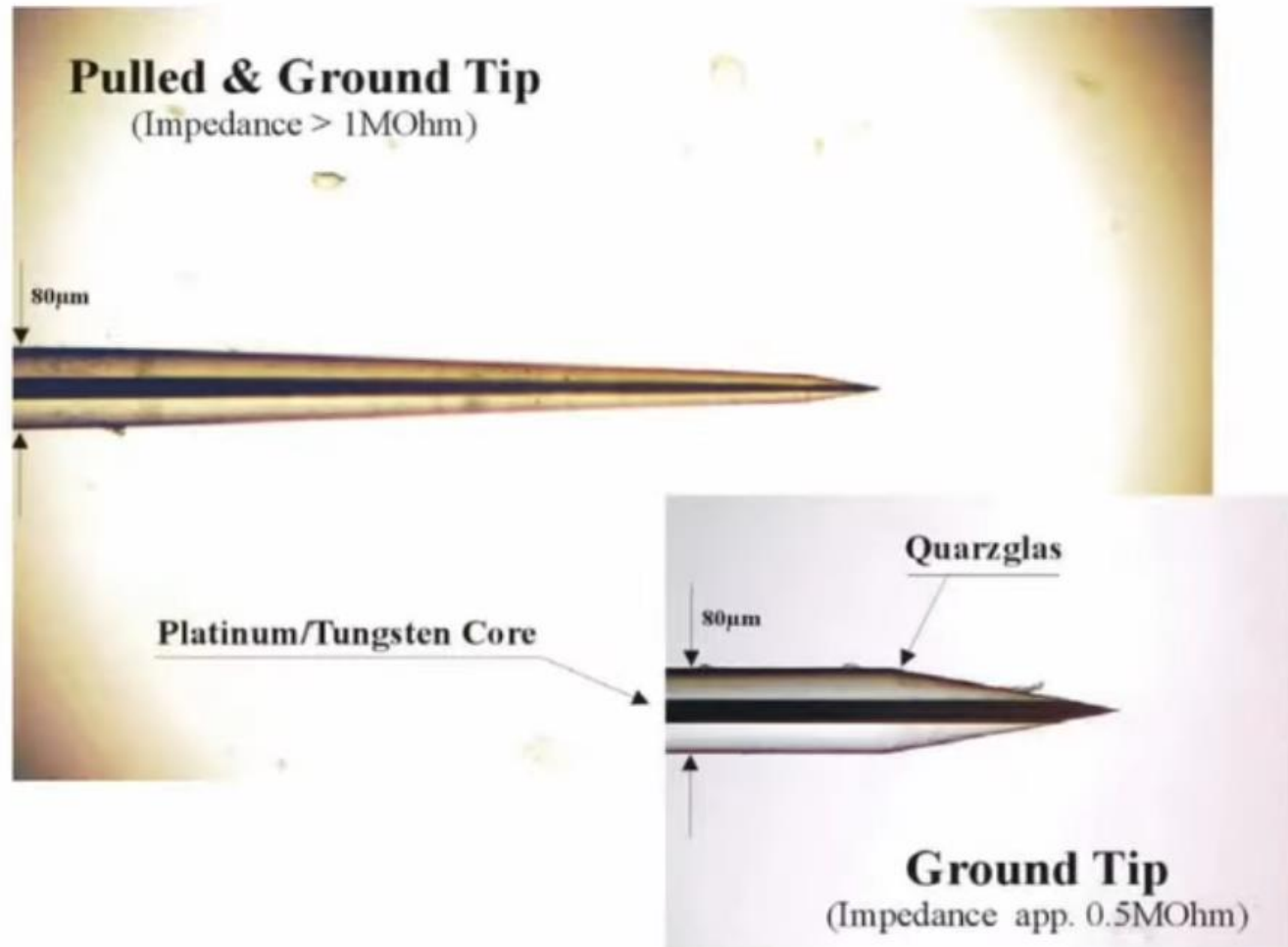
What is the origin of extracellular electrical signals ?

Methods for recording brain activity



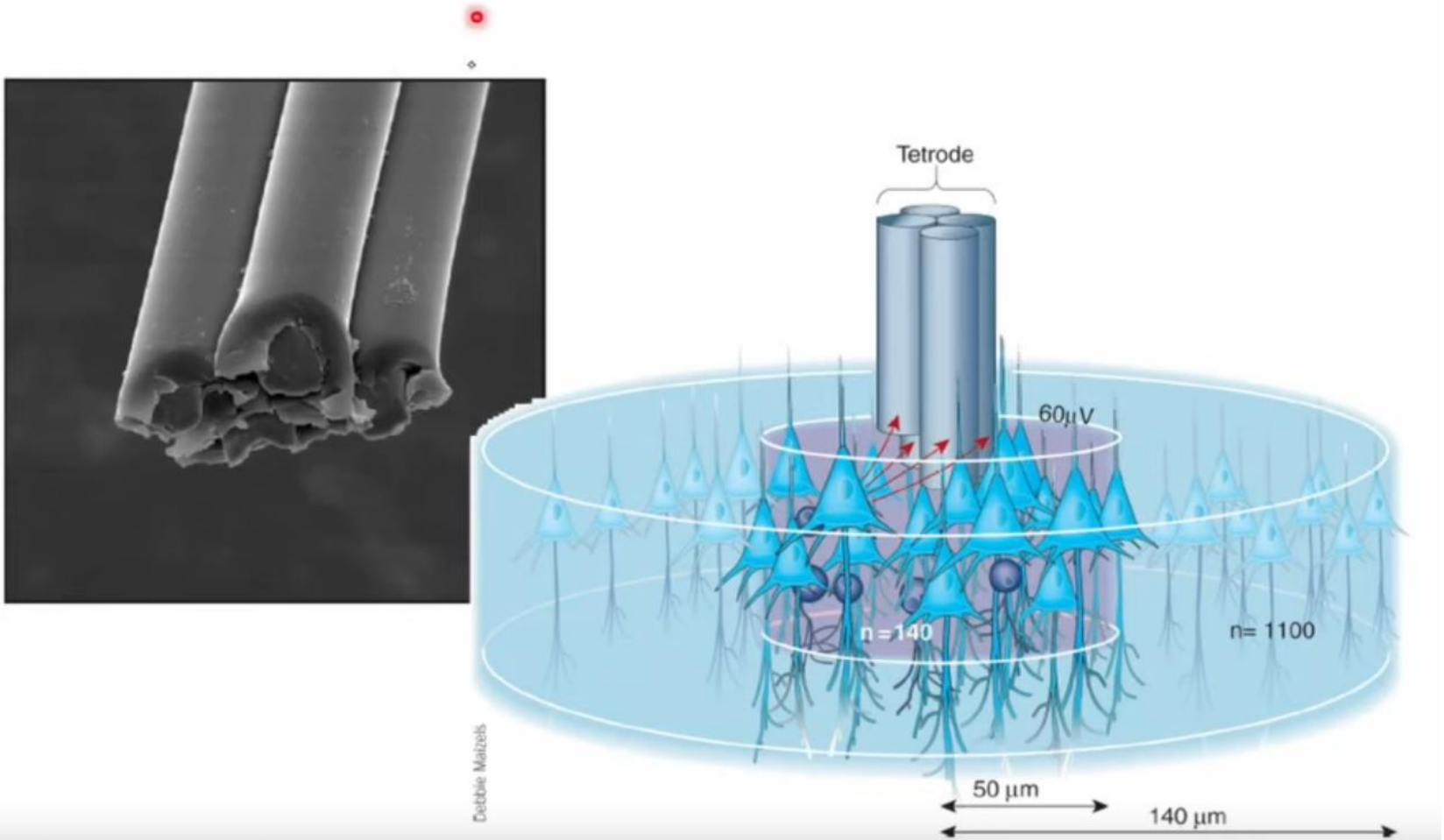
What is the origin of extracellular electrical signals ?

Tungsten microelectrodes



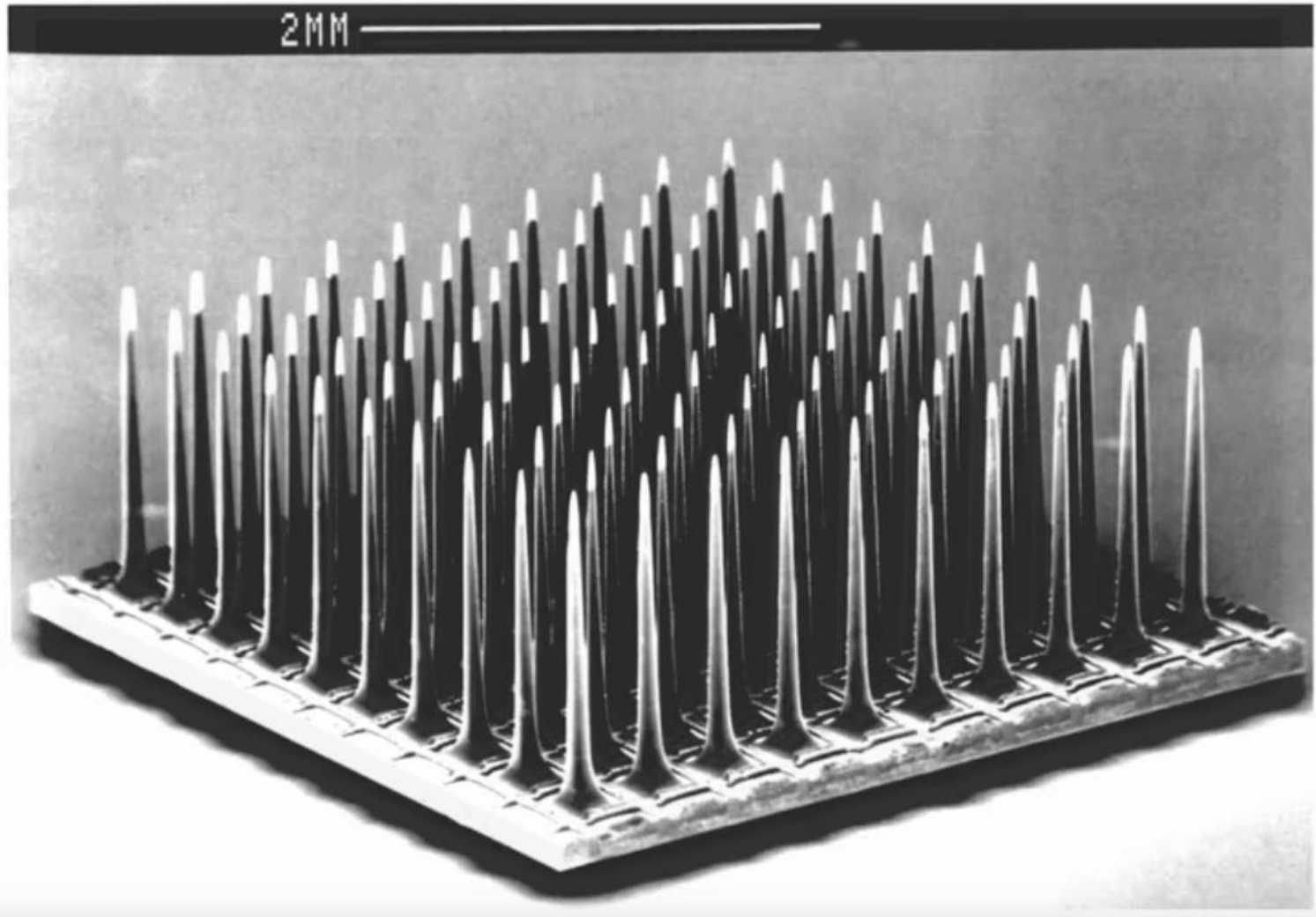
What is the origin of extracellular electrical signals ?

Tetrodes



What is the origin of extracellular electrical signals ?

Utah array

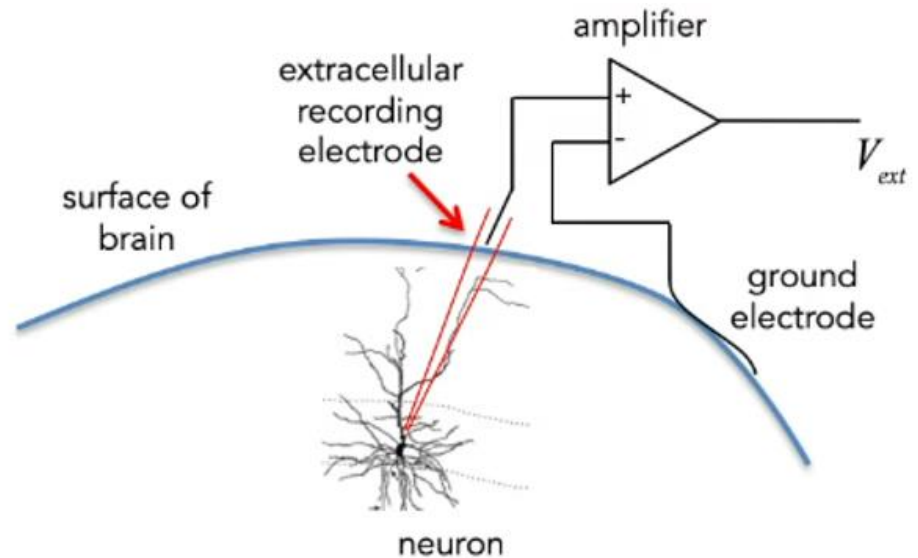


What is the origin of extracellular electrical signals ?

Extracellular voltages are always measured between the signal electrode and a local ground or reference electrode.

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Extracellular voltages are always measured between the signal electrode and a local ground or reference electrode.

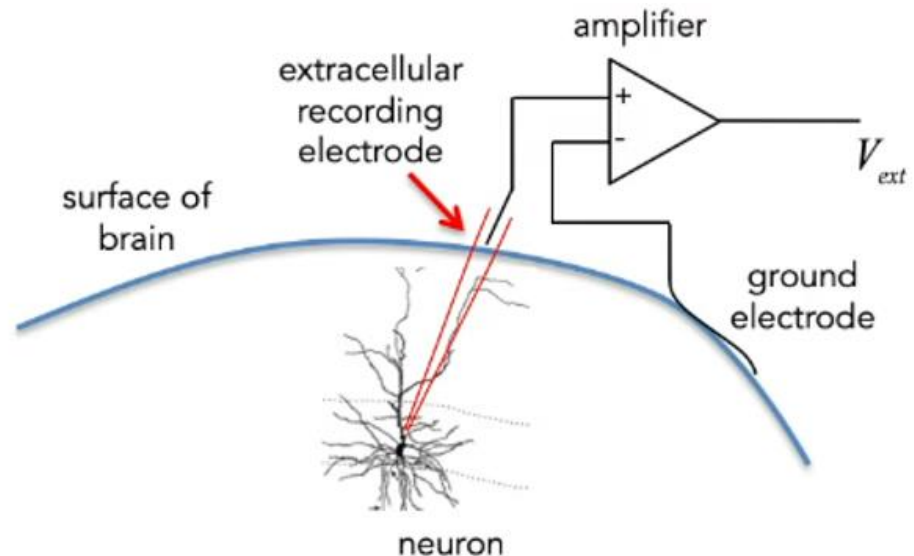


What is the origin of extracellular electrical signals ?

Extracellular voltages are always measured between the signal electrode and a local ground or reference electrode.

Extracellular voltages are always measured between the signal electrode and a local ground or reference electrode.

Voltage differences between the signal and ground electrode are always associated with current flow through the extracellular space.



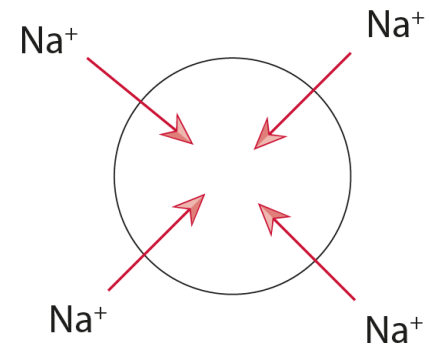
Back to Ohm's Law!

$$\Delta V = I R_{ext}$$

What is the origin of extracellular electrical signals ?

What happens to the voltage outside our model neuron during an action potential ?

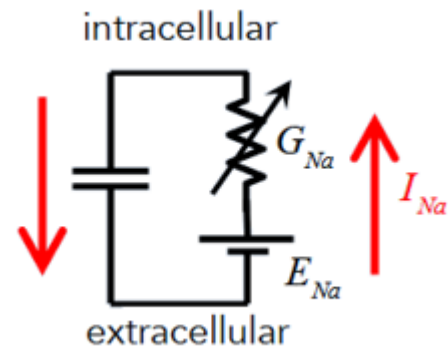
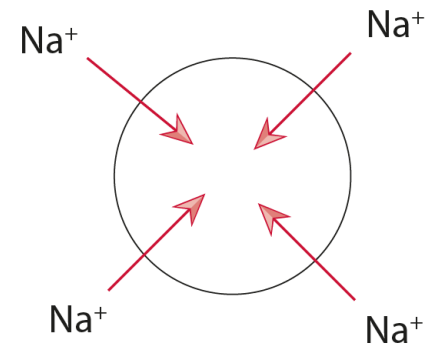
- Let's start with our spherical neuron



What is the origin of extracellular electrical signals ?

What happens to the voltage outside our model neuron during an action potential ?

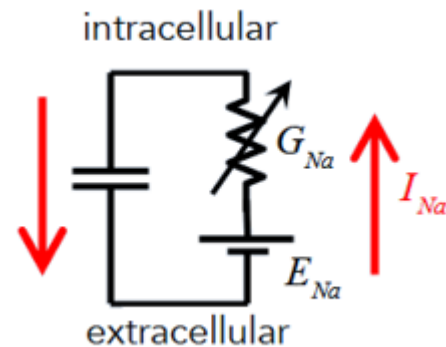
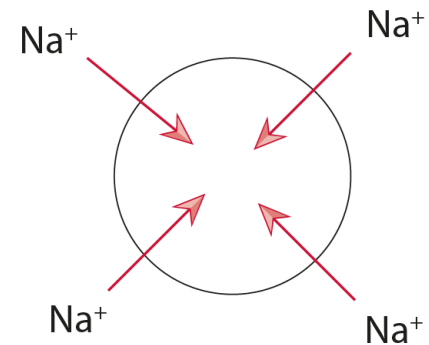
- Let's start with our spherical neuron
- There is no spatial separation between current flows into the neuron and current flow out of the neuron. Thus, no current flow outside the neuron.



What is the origin of extracellular electrical signals ?

What happens to the voltage outside our model neuron during an action potential ?

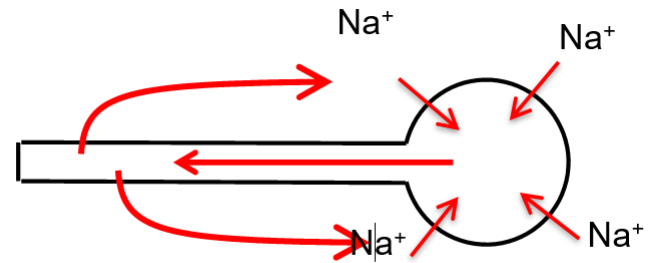
- Let's start with our spherical neuron
- There is no spatial separation between current flows into the neuron and current flow out of the neuron. Thus, no current flow outside the neuron.
- And thus no extracellular voltage changes



What is the origin of extracellular electrical signals ?

What happens to the voltage outside our model neuron during an action potential ?

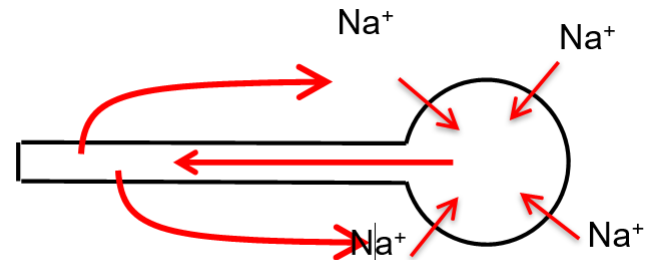
Now let's see what happens when we add a dendrite



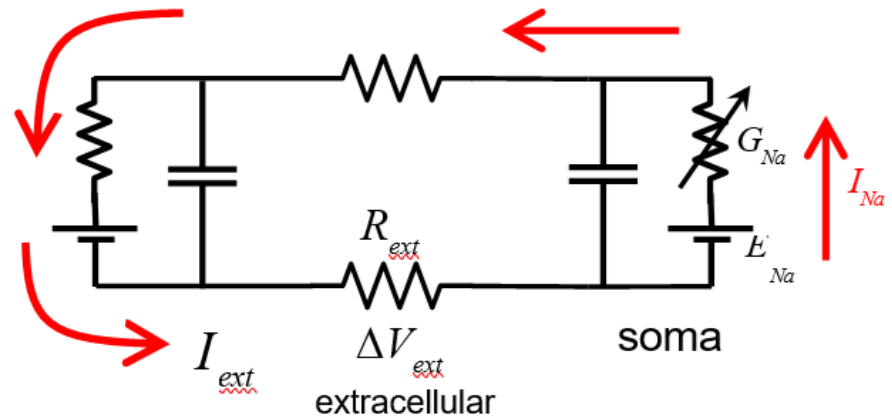
What is the origin of extracellular electrical signals ?

What happens to the voltage outside our model neuron during an action potential ?

Now let's see what happens when we add a dendrite



Now there is extracellular current flow

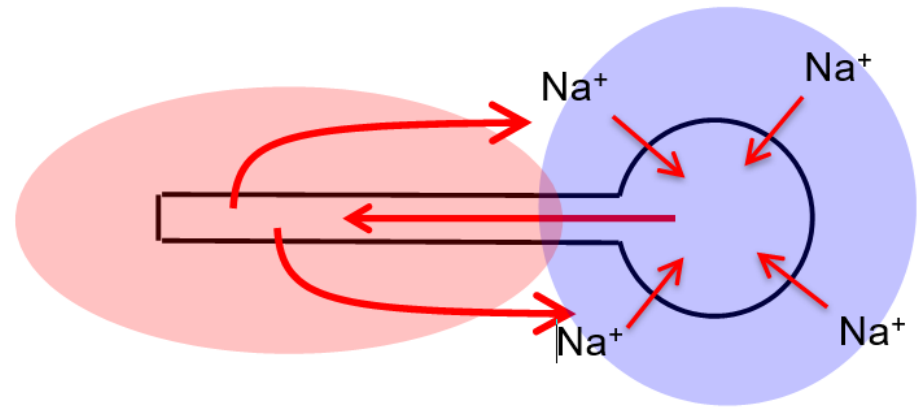


$$\Delta V_{\text{ext}} = R_{\text{ext}} I_{\text{ext}}$$

And extracellular voltage changes!

Current sources and sinks

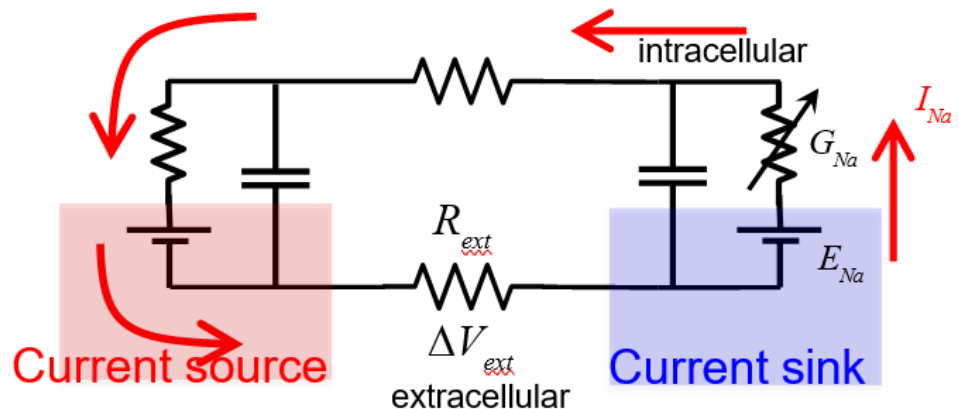
What happens to the voltage outside our model neuron during an action potential ?



Extracellular current flows from current sources to current sinks

Current sources are regions of higher extracellular potential

Current sinks are regions of lower extracellular potential

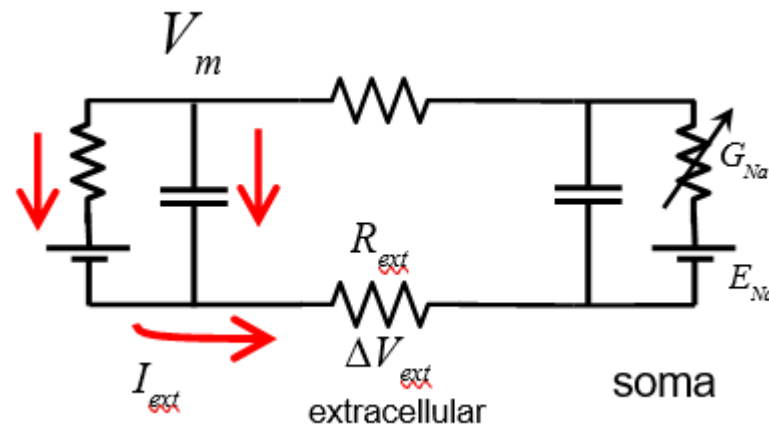


Relation between membrane potential and extracellular potential

$$\Delta V_{ext} = R_{ext} I_{ext} = R_{ext} (I_c + I_R)$$

$$= R_{ext} \left[C \frac{dV_m}{dt} + G(V_m - E_L) \right]$$

$$\Delta V_{ext} \approx R_{ext} C \frac{dV_m}{dt}$$

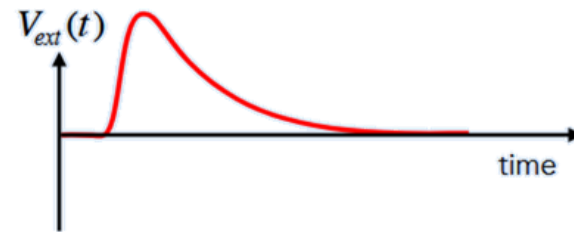
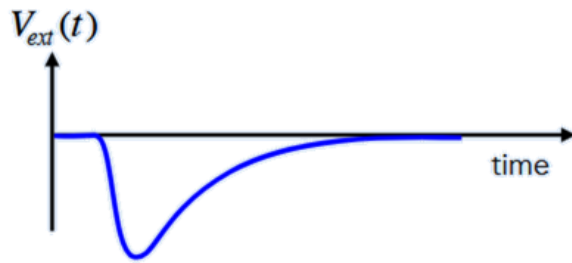
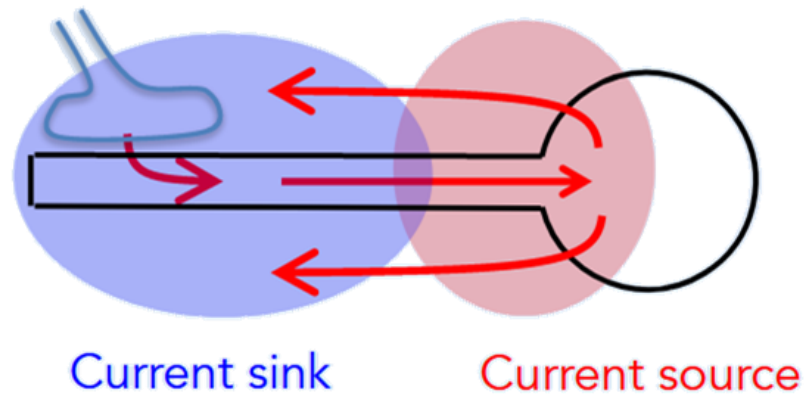


Extracellular voltages look a lot like the derivative on membrane potential!

Origin of extracellular signals ?

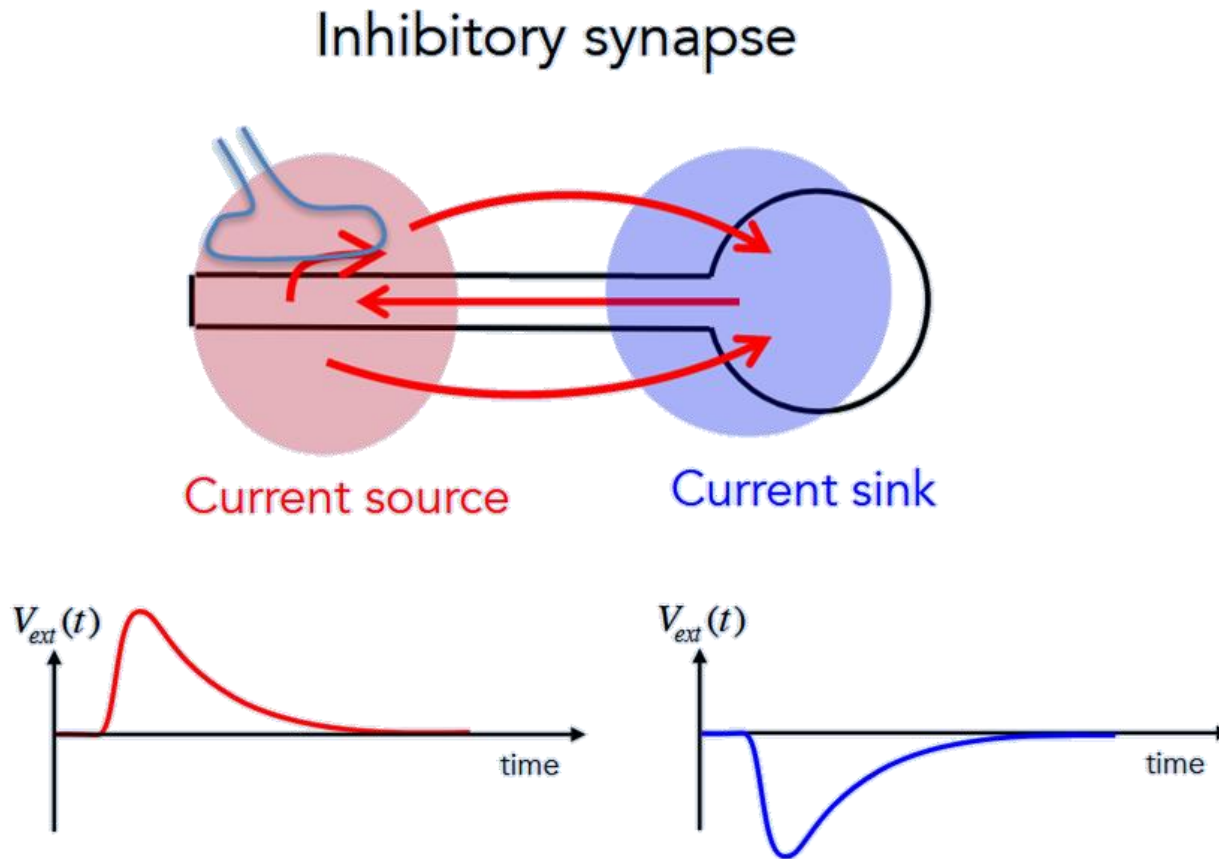
What happens to the voltage outside our model neuron during a synaptic input?

Excitatory synapse



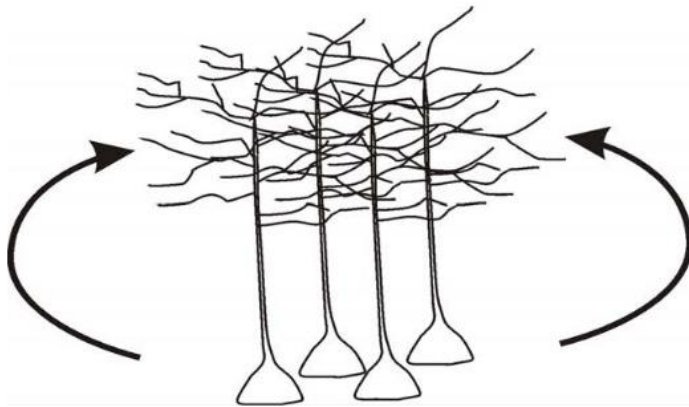
Origin of extracellular signals ?

What happens to the voltage outside our model neuron during a synaptic input?

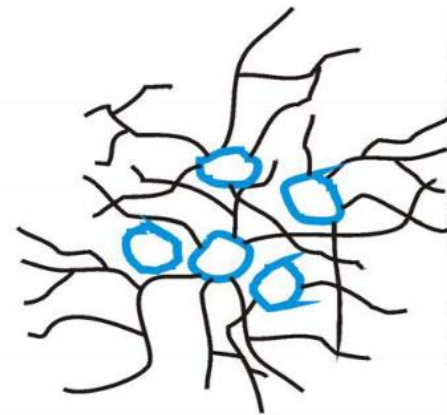


Local Field potentials

- Synchronous synaptic input to large populations of neurons
- Depends on linear summation
morphology is important

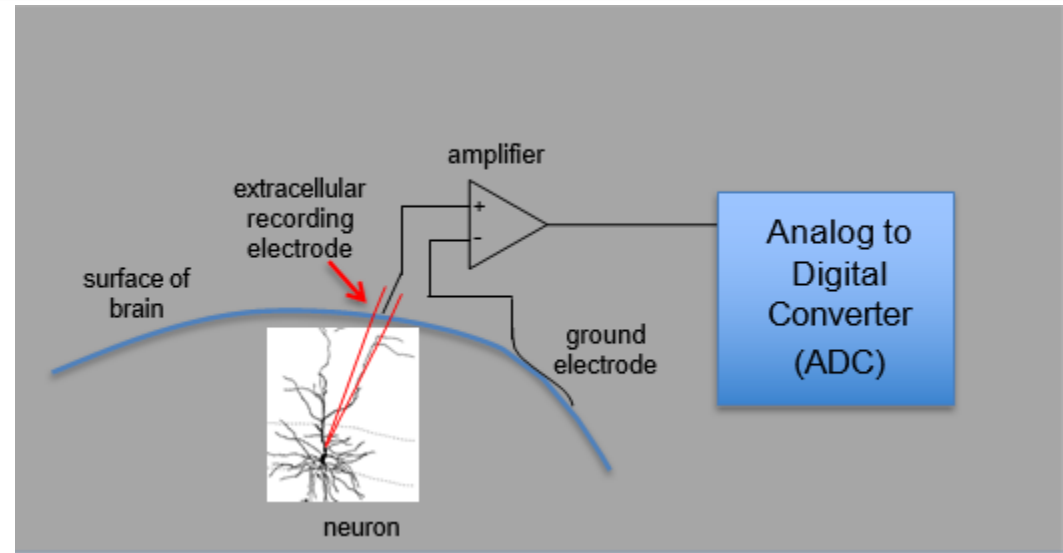


Laminar morphology
– large LFP



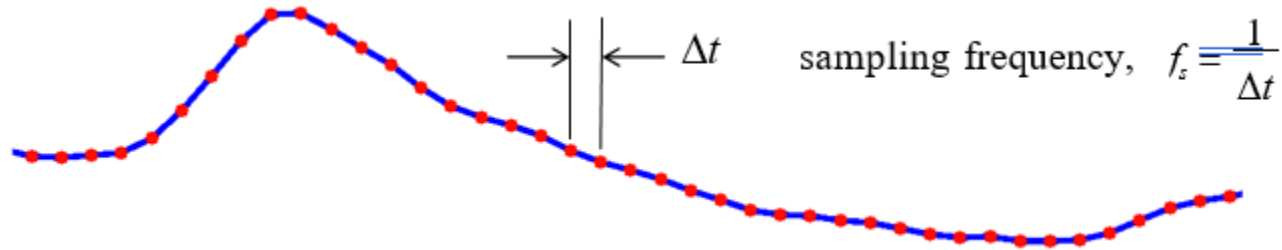
Nuclear morphology
– small LFP

Electrical recordings of the brain activity



SAMPLING

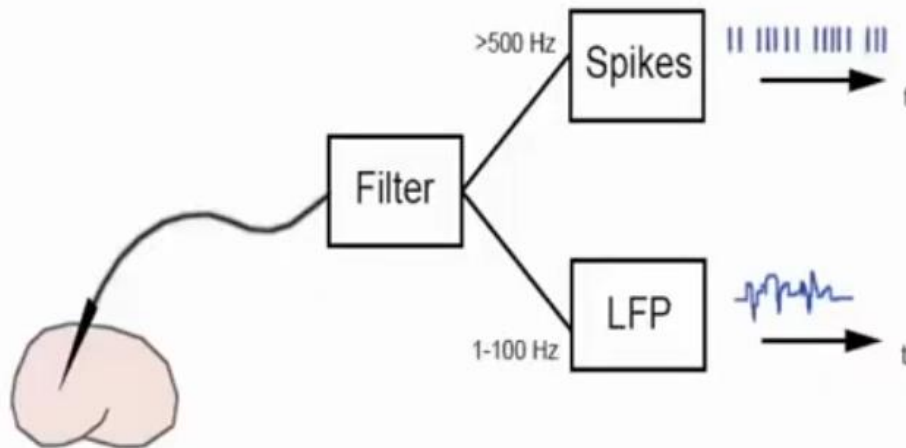
The analog to digital converter samples the voltage from the amplifier at regular intervals in time and stores the result in the computer memory.



The rate at which the samples are acquired is called the sampling rate or sampling frequency.

Two kinds of signals into the raw signal : Spikes and LFP

Typical analysis of extracellular recordings inside brain

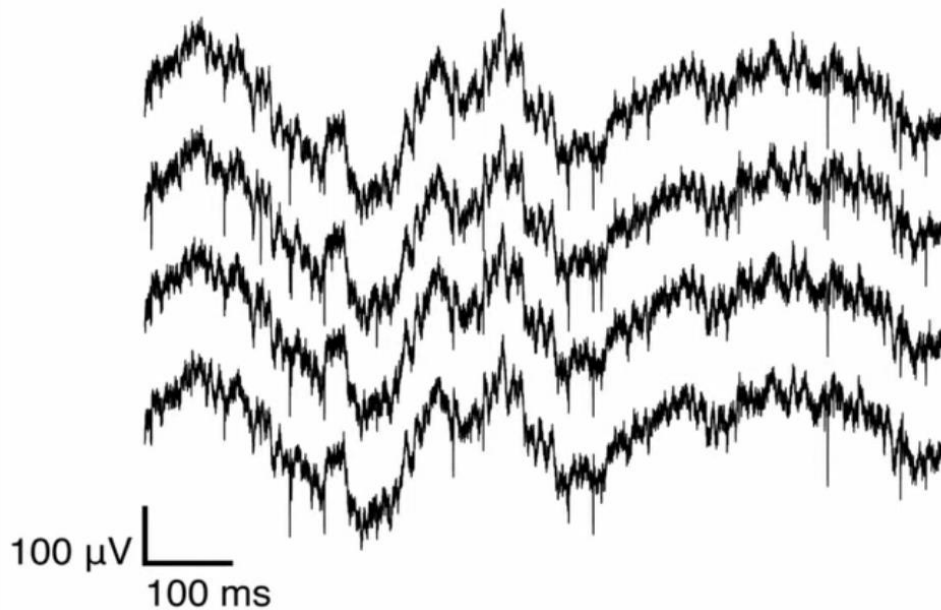


Recorded signal split into two frequency bands:

- High-frequency band (>500 Hz), **Multi-unit activity (MUA)**
 - measures spikes in individual neurons around electrode tip
- Low-frequency band (<300 Hz): **Local field potential (LFP)**
 - measures subthreshold activity due to synaptic inputs from populations of neurons

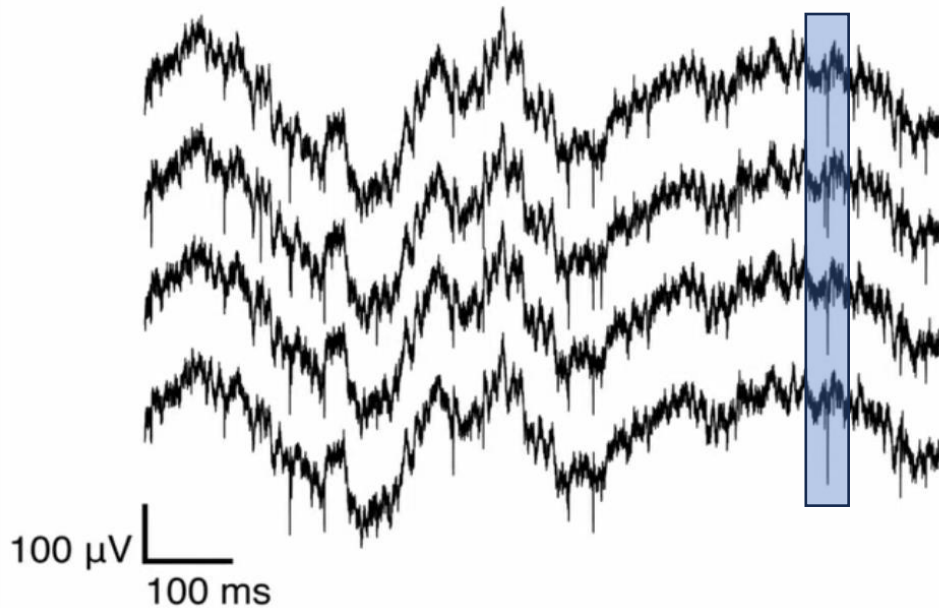
Two kinds of signals into the raw signal : Spikes and LFP

Raw signal



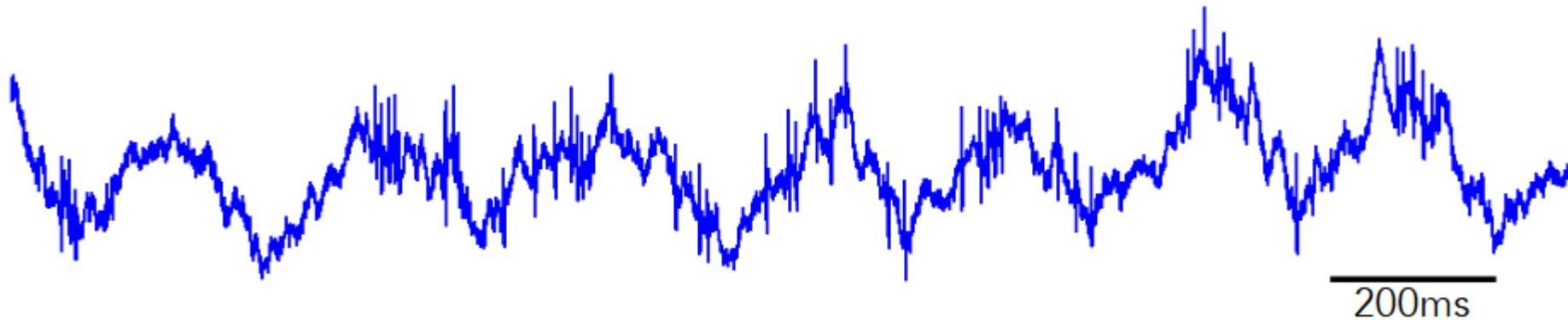
Two kinds of signals into the raw signal : Spikes and LFP

Raw signal

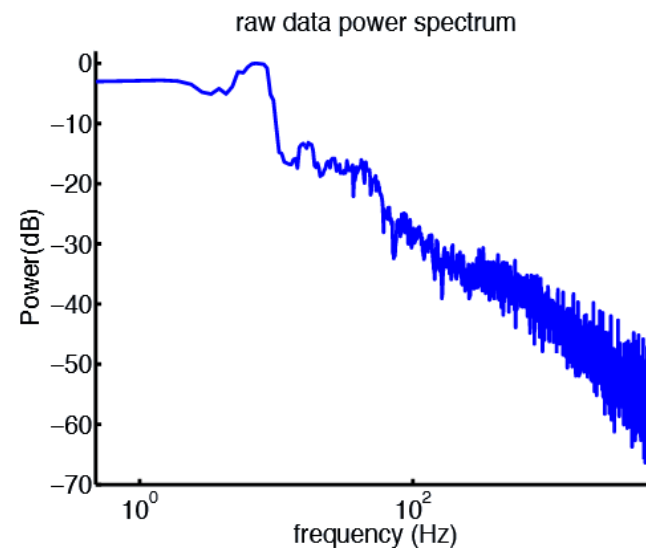
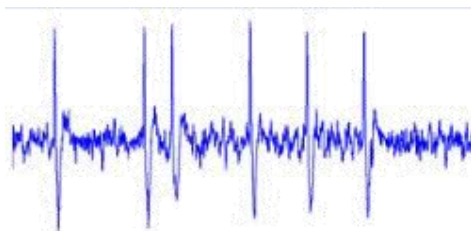


Two kinds of signals into the raw signal : Spikes and LFP

- Recording of neural activity in hippocampus of a running rat

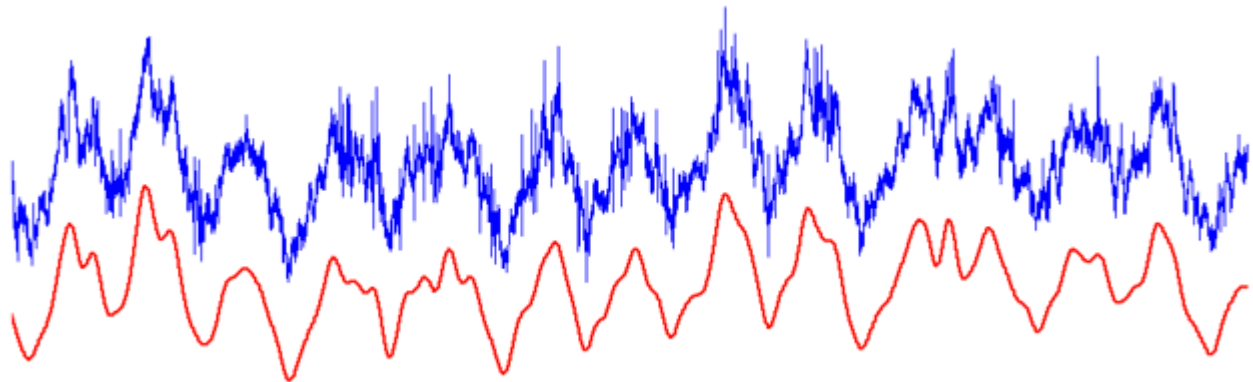
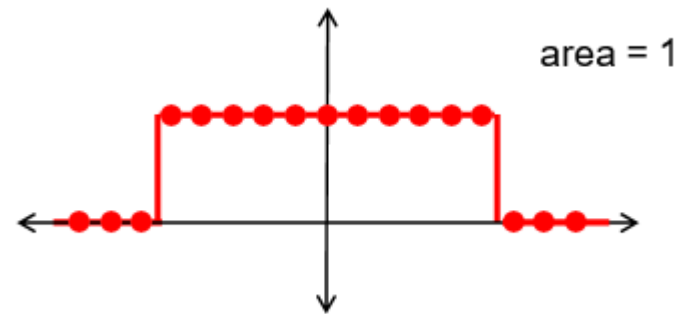
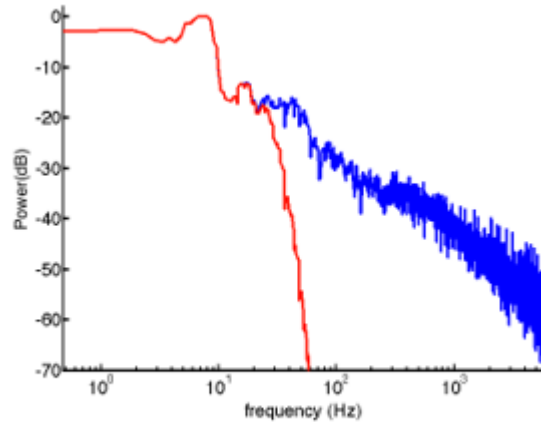


- Slow – Local Field Potentials (LFP)
Synaptic currents
- Fast – spikes
Action potentials



Low-pass filtering

Low-pass filtering can be done by convolving the signal with a kernel like this.

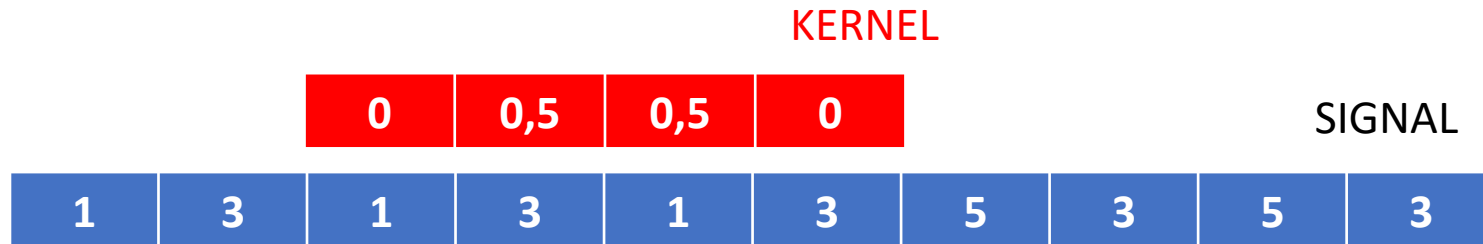


Explanation of low pass filter : extraction of LFP

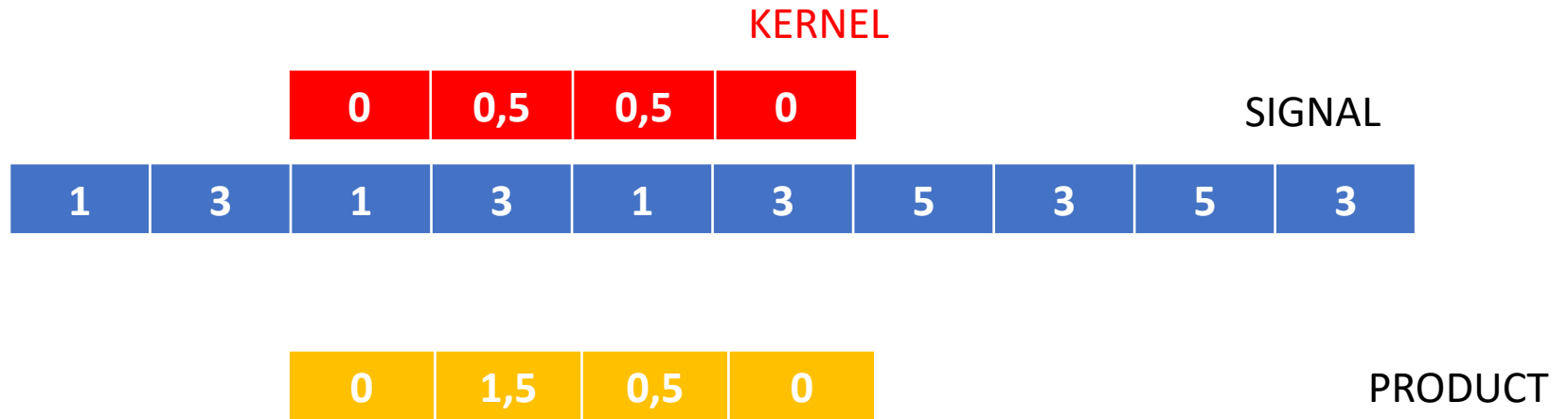
KERNEL

0	0,5	0,5	0
---	-----	-----	---

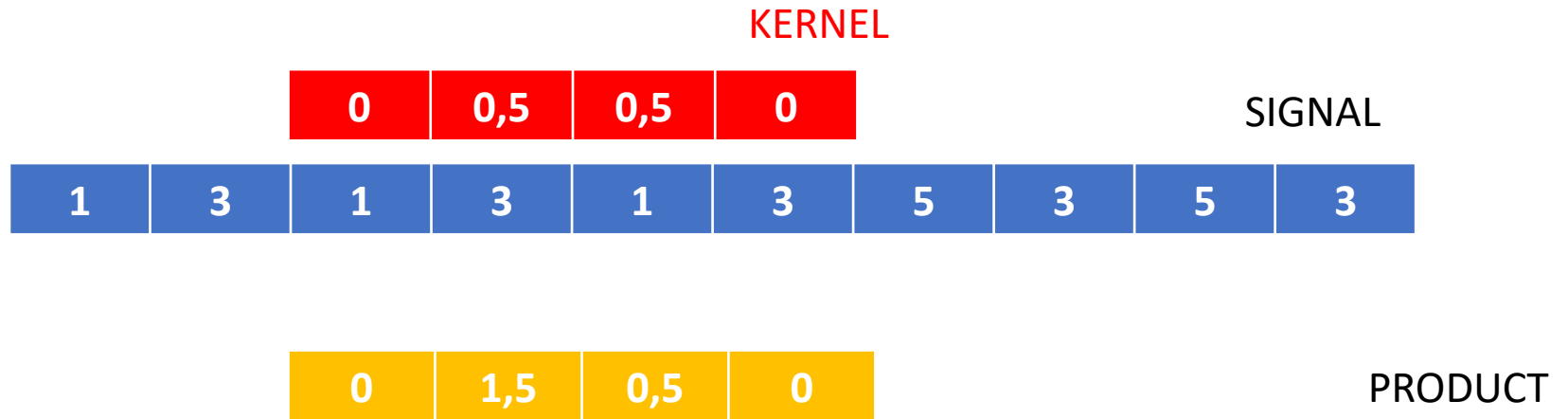
Explanation of low pass filter : extraction of LFP



Explanation of low pass filter : extraction of LFP

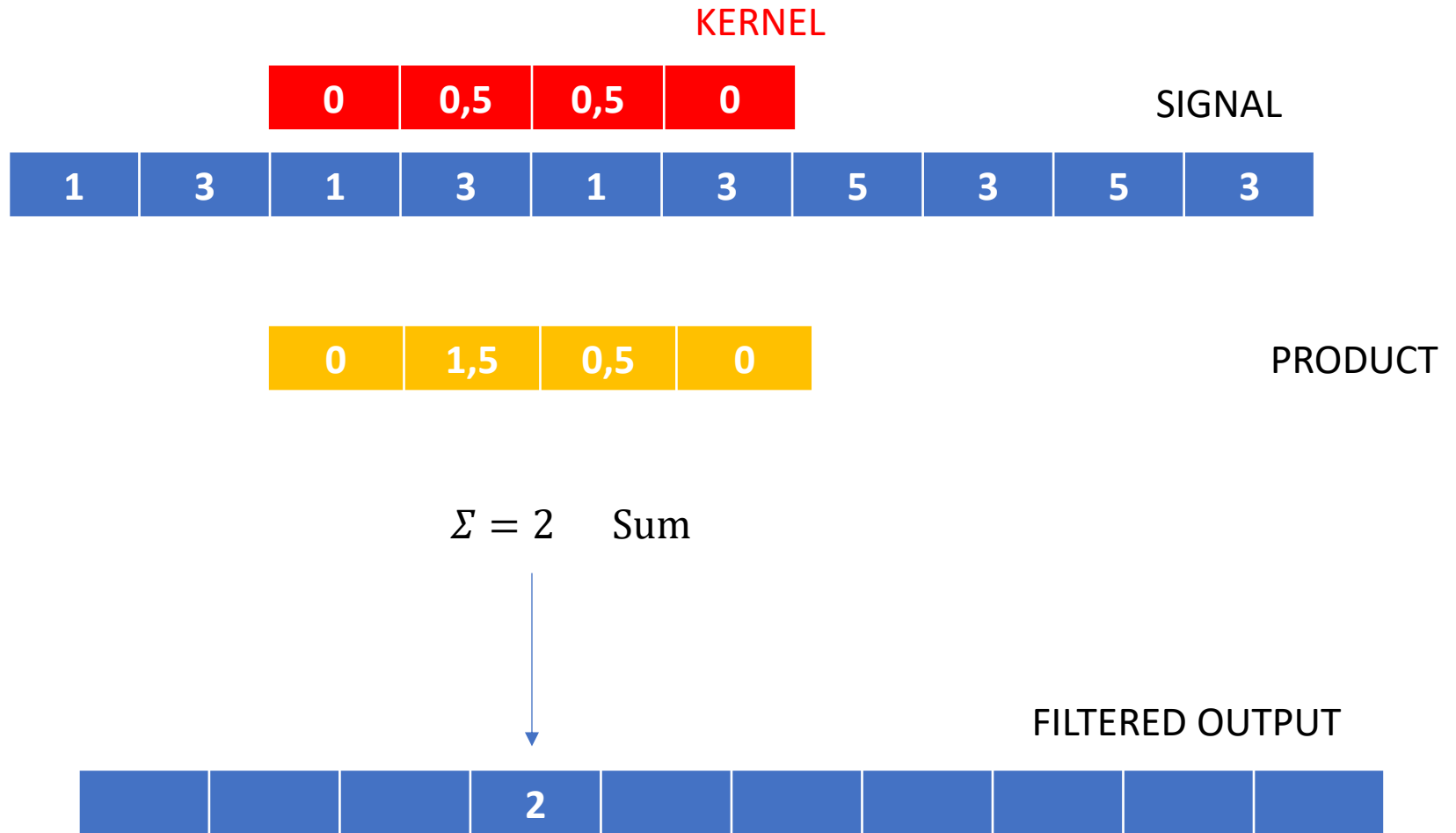


Explanation of low pass filter : extraction of LFP

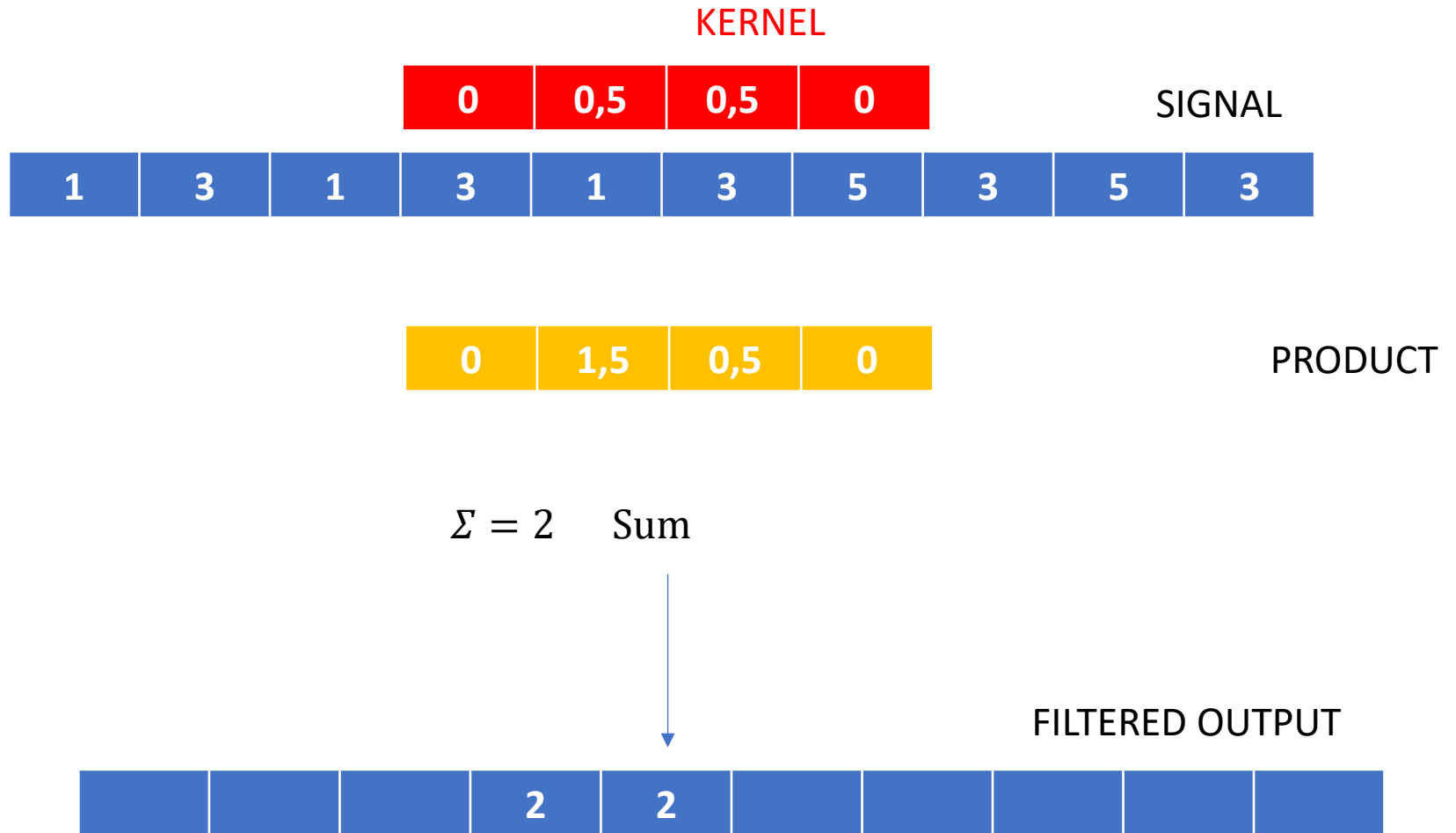


$$\Sigma = 2 \quad \text{Sum}$$

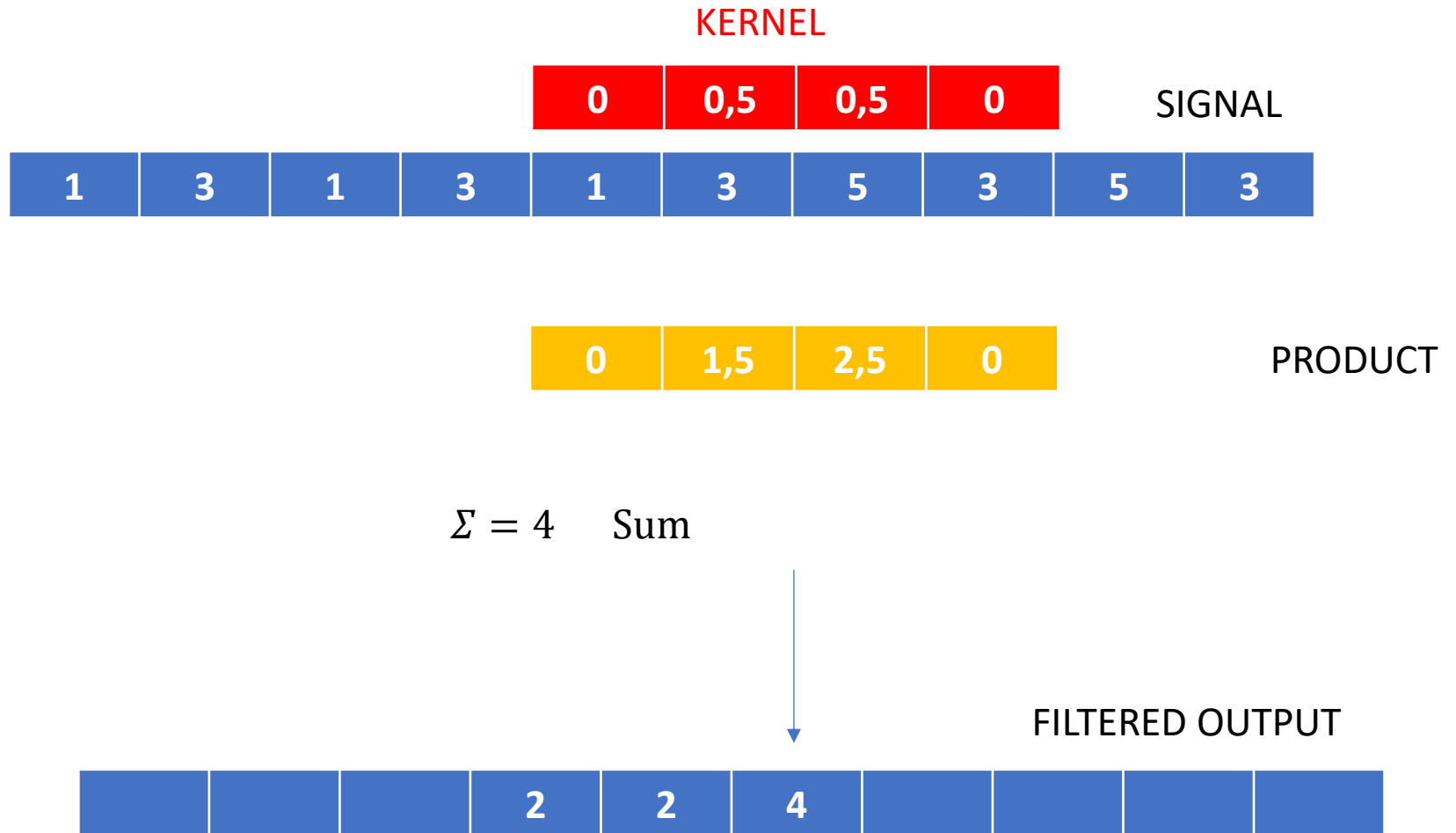
Explanation of low pass filter : extraction of LFP



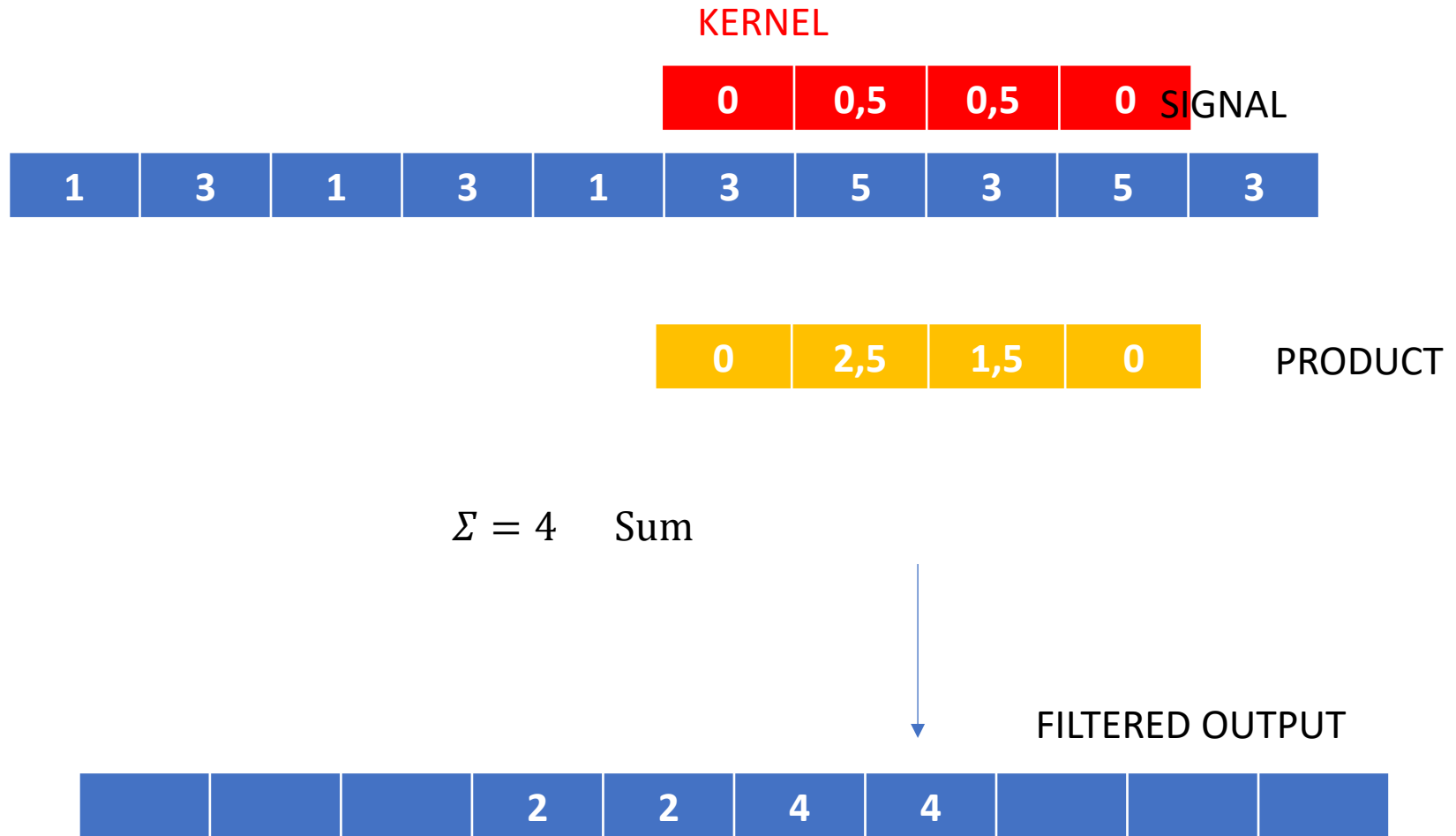
Explanation of low pass filter : extraction of LFP



Explanation of low pass filter : extraction of LFP



Explanation of low pass filter : extraction of LFP



Explanation of low pass filter : extraction of LFP

KERNEL

SIGNAL

1	3	1	3	1	3	5	3	5	3
---	---	---	---	---	---	---	---	---	---

0	2,5	1,5	0
---	-----	-----	---

PRODUCT

$\Sigma = 4$ Sum

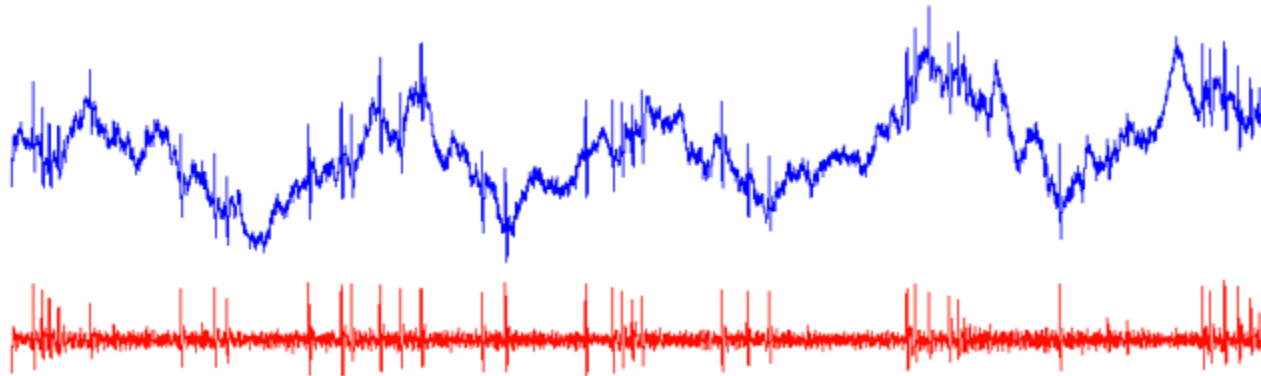
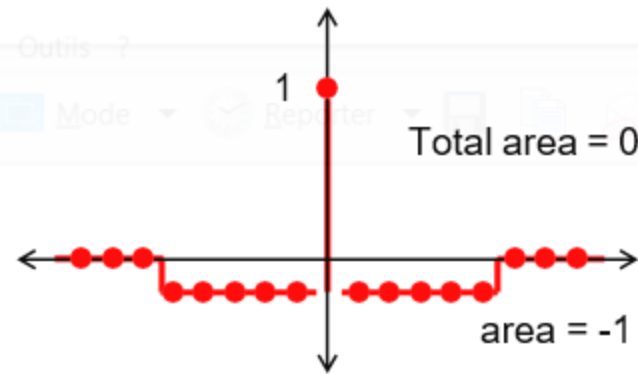
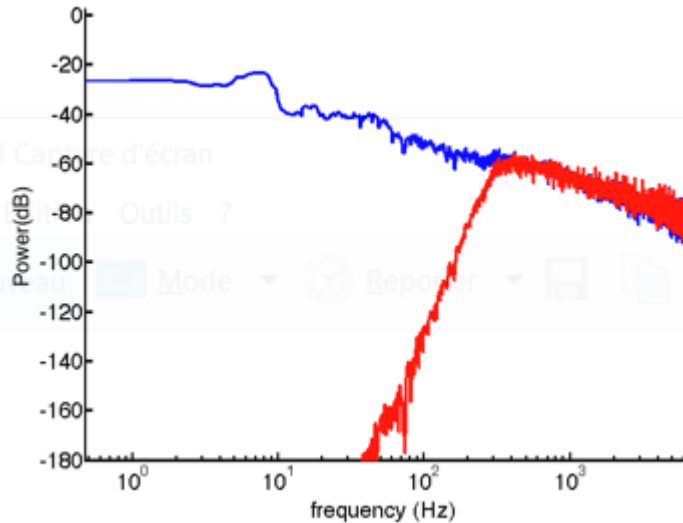


FILTERED OUTPUT

2	2	2	2	2	4	4	4	4	4
---	---	---	---	---	---	---	---	---	---

High-pass filtering : extraction of spikes

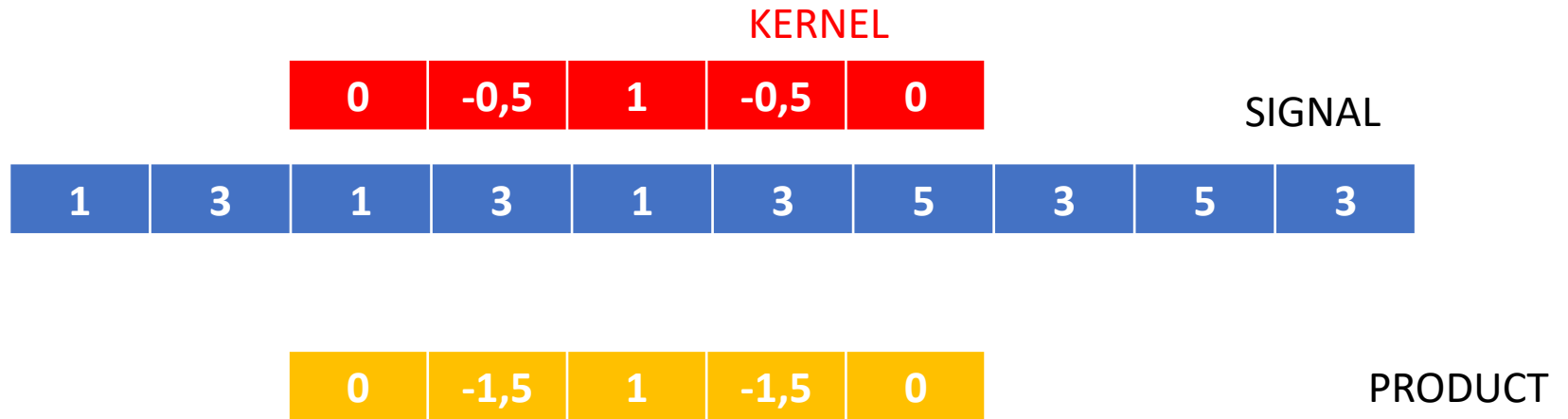
High-pass filtering can be done by convolving the signal with a kernel like this



High-pass filtering : extraction of spikes

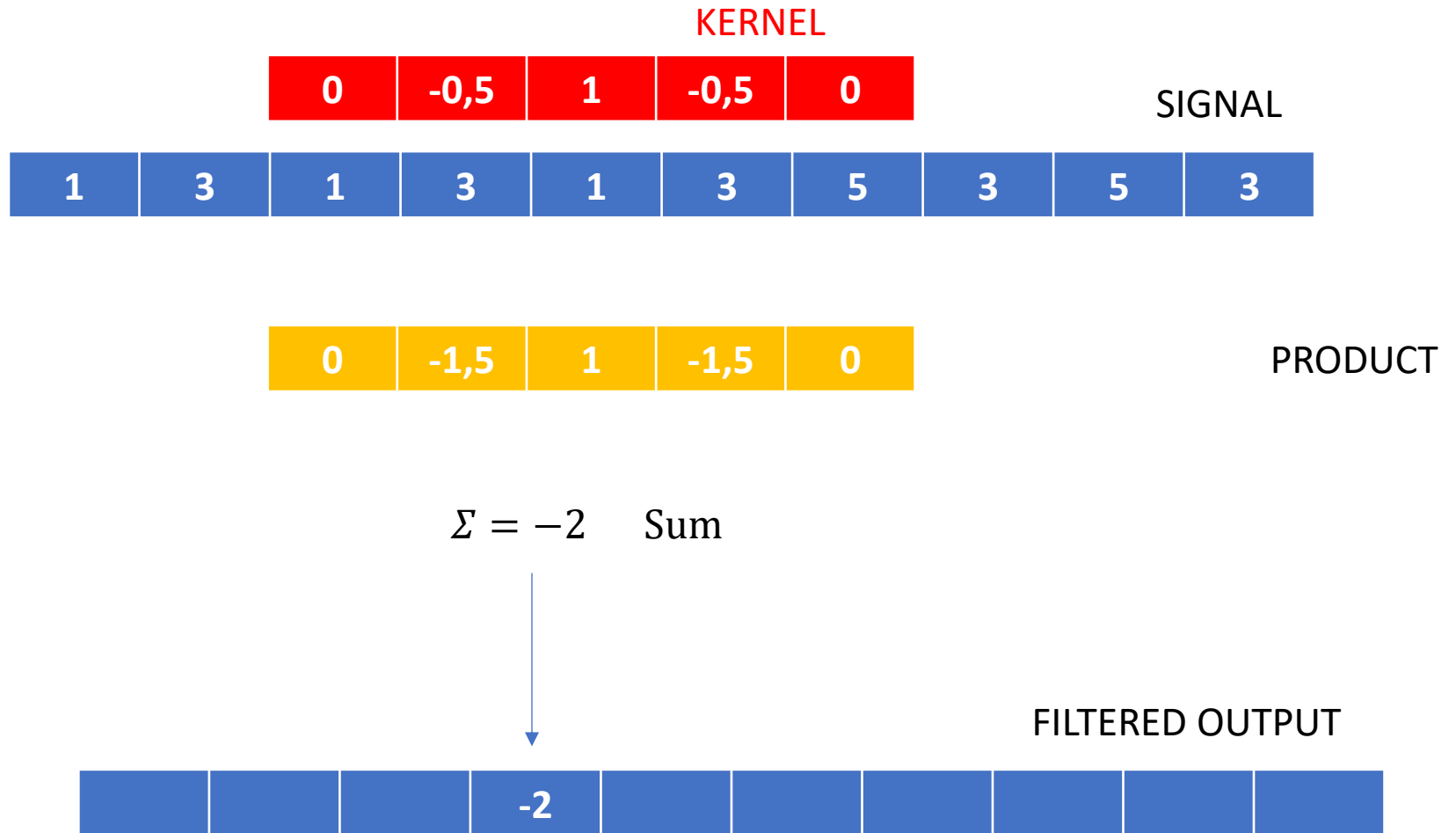
KERNEL					SIGNAL				
0	-0,5	1	-0,5	0					
1	3	1	3	1	3	5	3	5	3
0	-1,5	1	-1,5	0					
					PRODUCT				

High-pass filtering : extraction of spikes

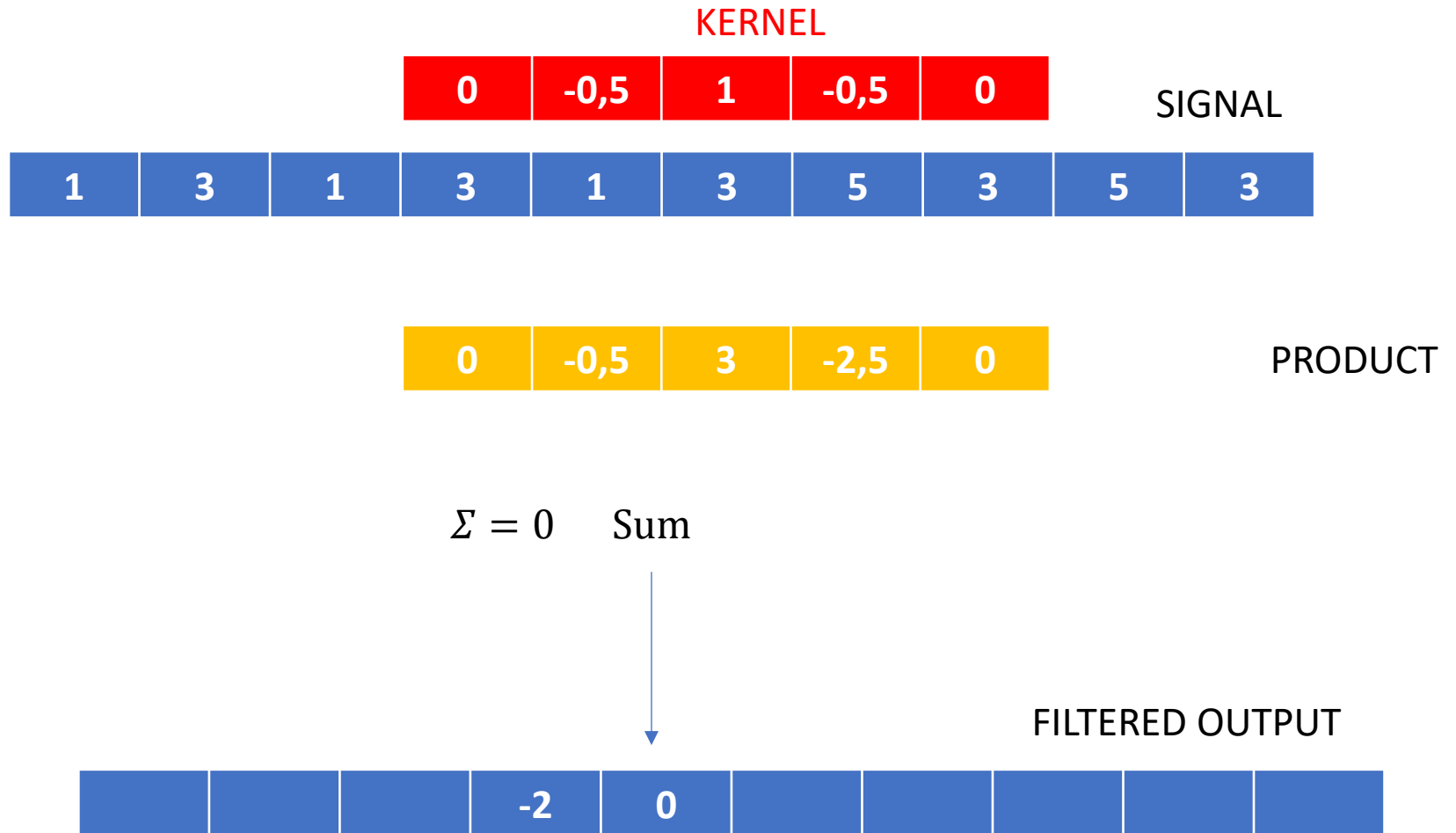


$$\Sigma = -2 \quad \text{Sum}$$

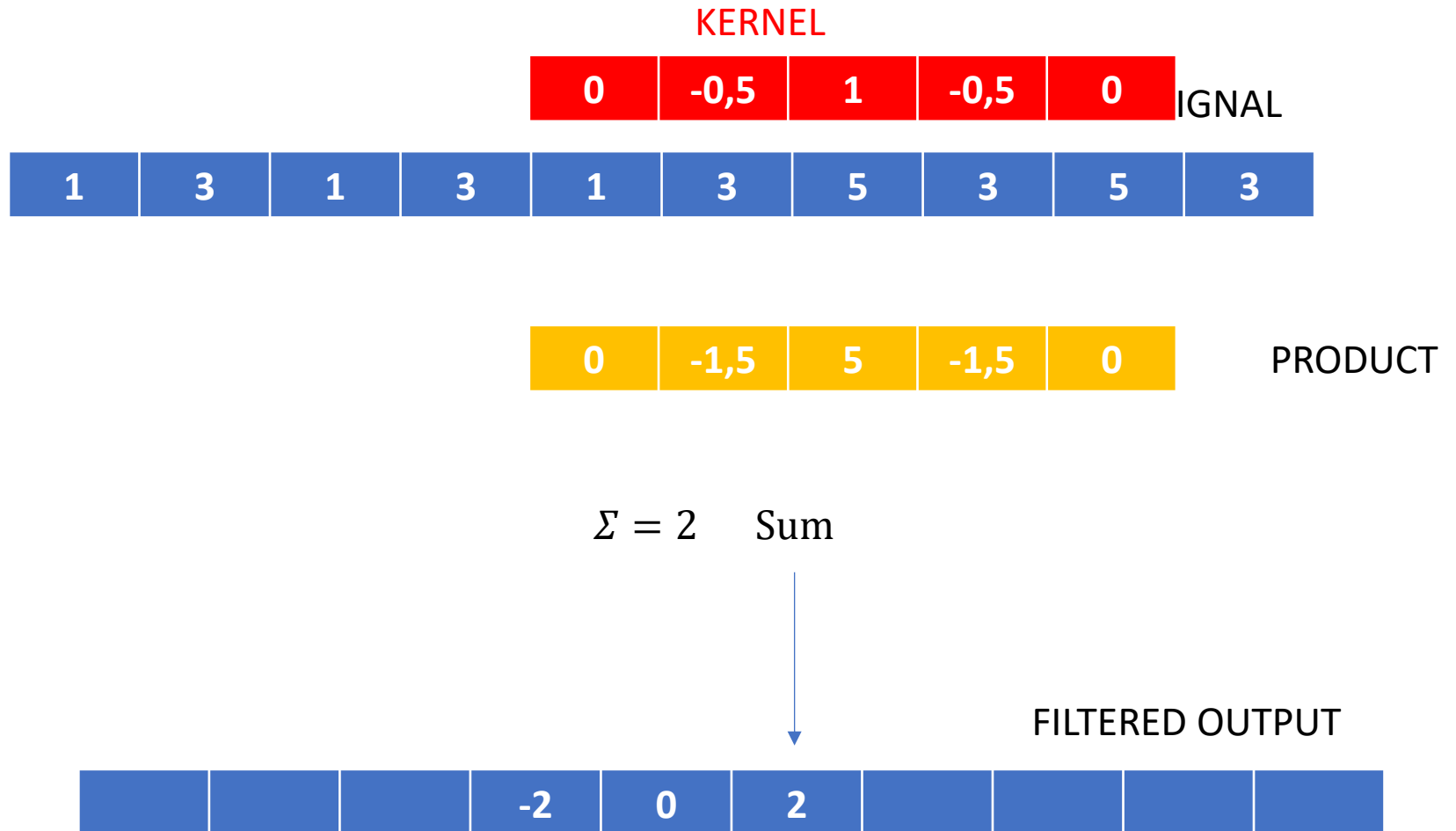
High-pass filtering : extraction of spikes



High-pass filtering : extraction of spikes



High-pass filtering : extraction of spikes



High-pass filtering : extraction of spikes

KERNEL

0	-0,5	1	-0,5	0
---	------	---	------	---

1	3	1	3	1	3	5	3	5	3
---	---	---	---	---	---	---	---	---	---

0	-2,5	3	-2,5	0
---	------	---	------	---

PRODUCT

$\Sigma = -2$ Sum



FILTERED OUTPUT

			-2	0	2	-2			
--	--	--	----	---	---	----	--	--	--

High-pass filtering : extraction of spikes

KERNEL

0	-0,5	1	-0,5	0
---	------	---	------	---

1	3	1	3	1	3	5	3	5	3
---	---	---	---	---	---	---	---	---	---

0	-2,5	3	-2,5	0
---	------	---	------	---

$\Sigma = -2$ Sum

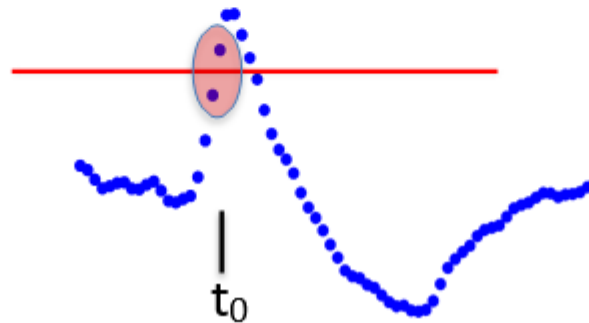
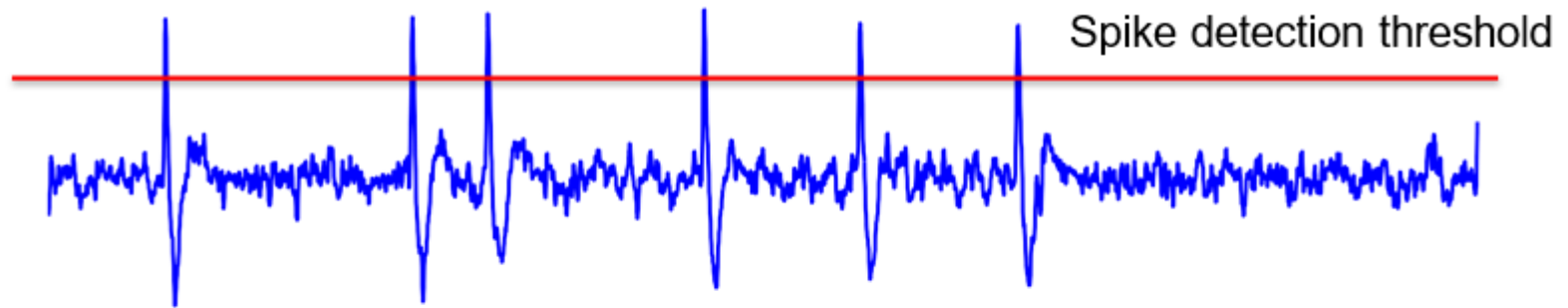


FILTERED OUTPUT

			-2	0	2	-2	2		
--	--	--	----	---	---	----	---	--	--

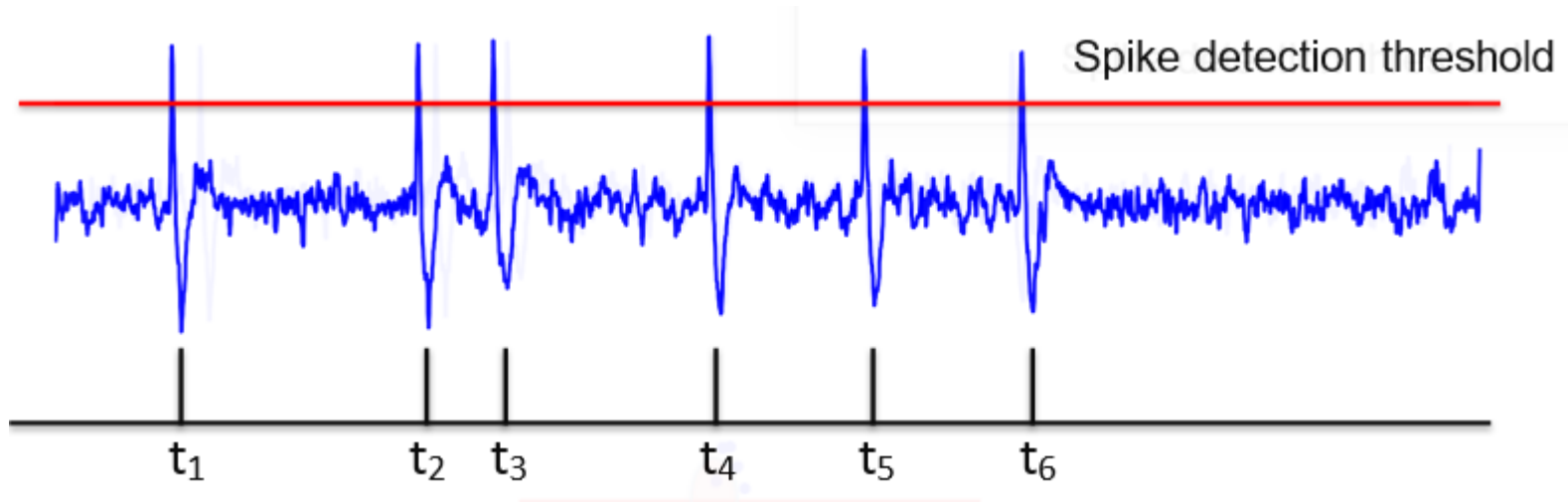
Spike detection

- If you are recording from only a single neuron, it is easy to extract spike times...



Spike trains

- Representation of all the spikes that a neuron generates in a period of time of interest

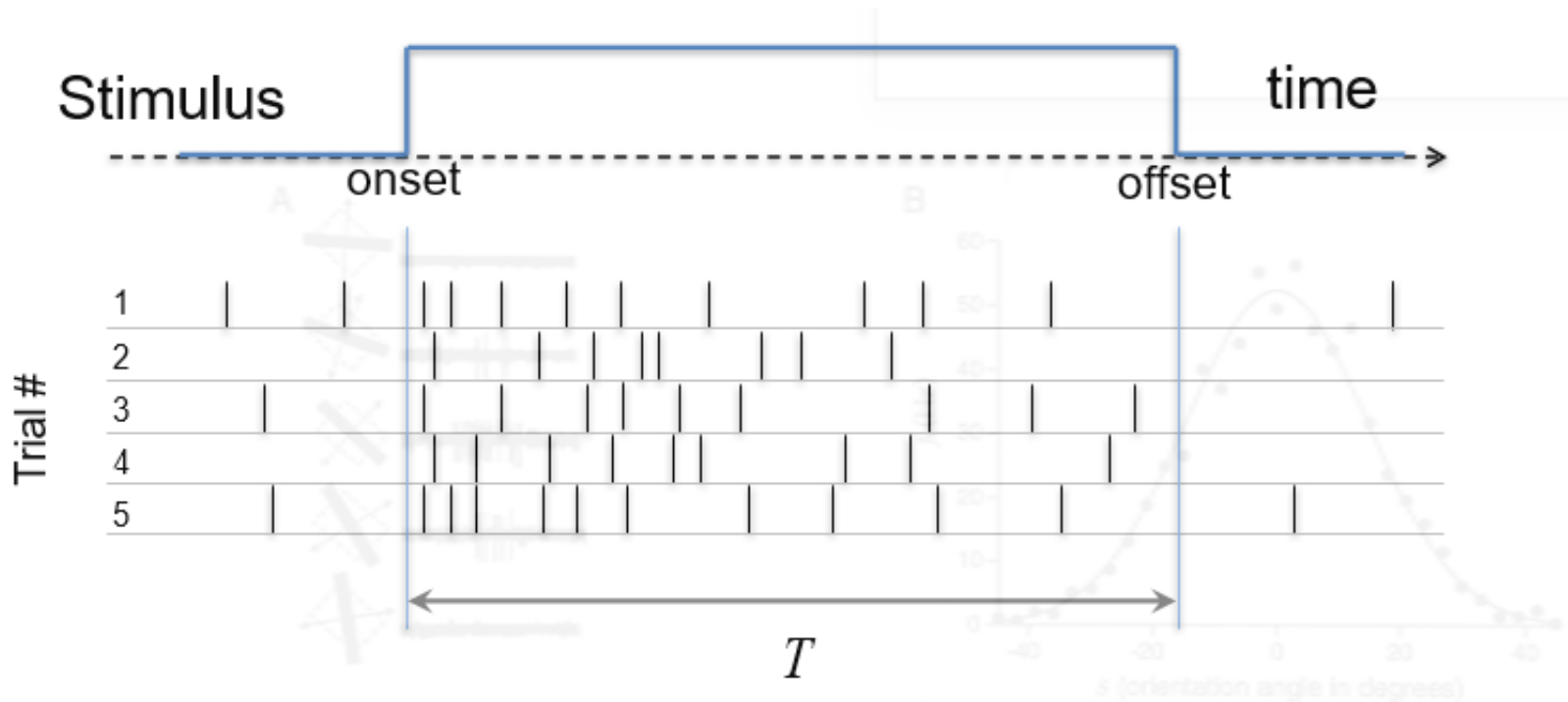


- We can represent a spike train as a list of spike times**

t_i for $i=1$ to N spikes

Quantifying firing rates

Method 1: Trial average firing rate

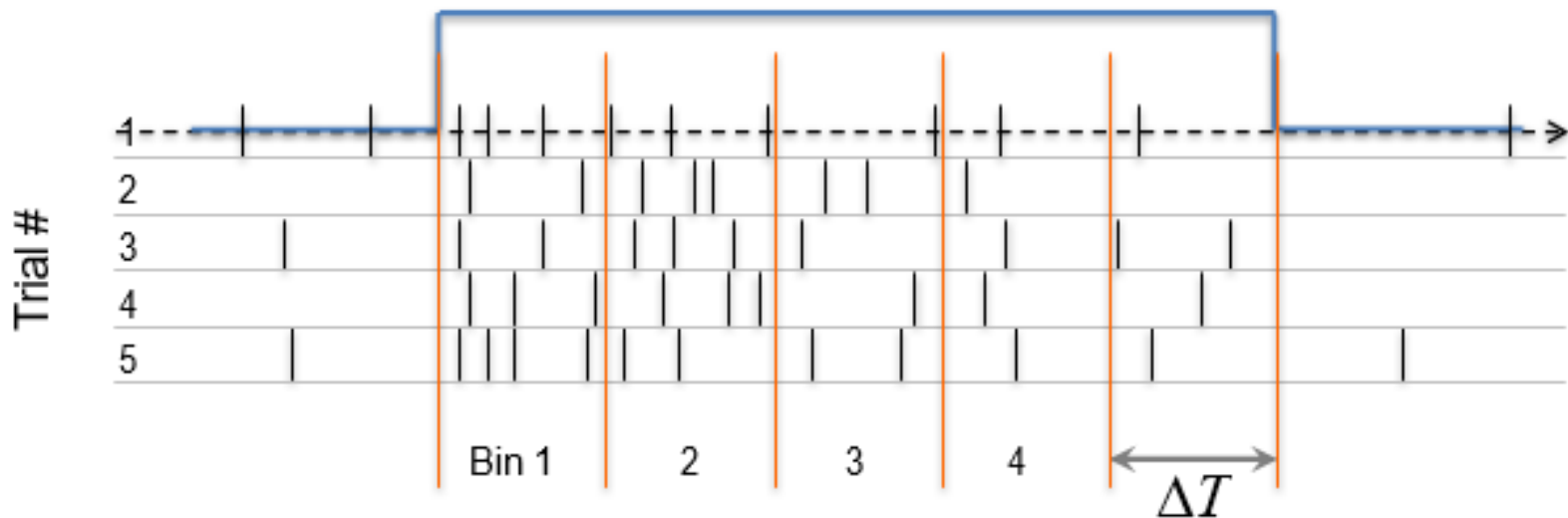


$$R = \frac{\langle N_i \rangle_i}{T}$$

N_i = number of spikes on trial i
 $\langle \rangle_i$ denotes the average over all trials i

Quantifying firing rates

We can get higher temporal resolution by breaking the rate calculation into smaller units in time



Trial-average rate in time bin j

$$R_j = \frac{1}{\Delta T} \langle N_{i,j} \rangle_i$$

N_i = number of spikes on trial i
 $\langle \rangle_i$ denotes the average over all trials i

Quantifying firing rates

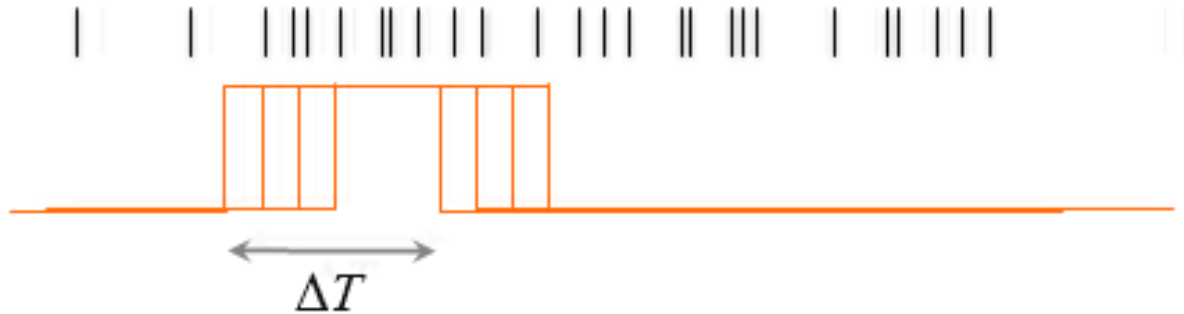
**Peri-Stimulus Time Histogram
(PSTH)**

or

**Peri-Event Time Histogram
(PETH)**

Quantifying firing rates

- A continuous measure of firing rate.

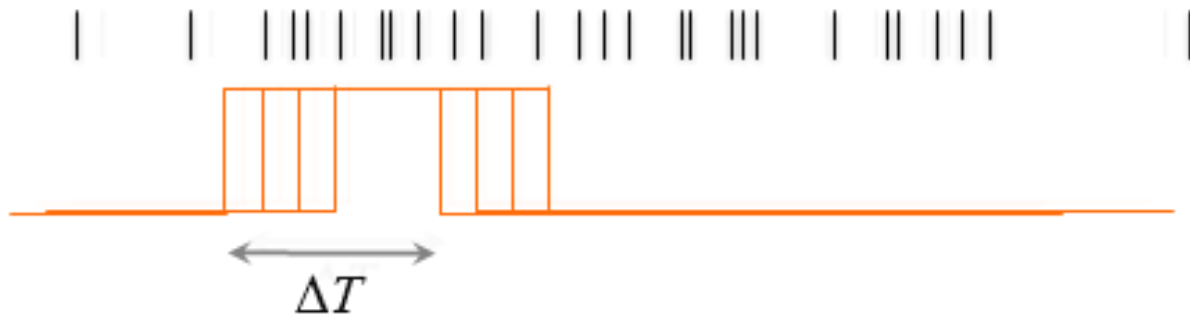


We count the number of spikes in a small window of width ΔT and shift the window in smaller steps

**How can we describe this mathematically
(you ask)?**

Quantifying firing rates

- A continuous measure of firing rate.

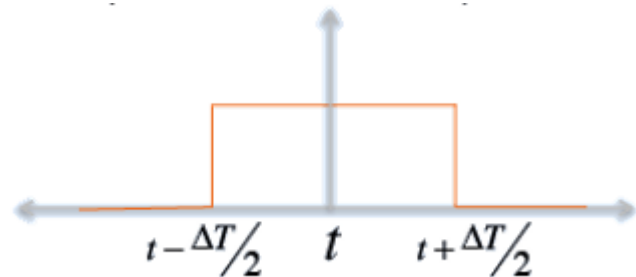


We can write this process down mathematically as follows.

First, we use the fact that the number of spikes in an interval t_1 to t_2 is given by

$$N = \int_{t_1}^{t_2} \rho(t) dt$$

$$R(t) = \frac{N}{\Delta T} = \frac{1}{\Delta T} \int_{t-\Delta T/2}^{t+\Delta T/2} \rho(\tau) d\tau$$



Quantifying firing rates

- A continuous measure of firing rate.



Even better is to convolve with a Gaussian kernel

$$\underline{R(t)} = \rho(t) * K$$

Kernel

$$K(\tau) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\tau^2}{2\sigma^2}}$$

The figure includes a graph of the Gaussian kernel function $K(\tau)$. The curve is a red bell shape centered at the origin of a coordinate system. The horizontal axis is labeled τ and the vertical axis is labeled $K(\tau)$. The area under the curve is indicated as 'area = 1'.

Gaussian kernel is still averaging - it is just a weighted average, with less weight at the edges

Quantifying firing rates

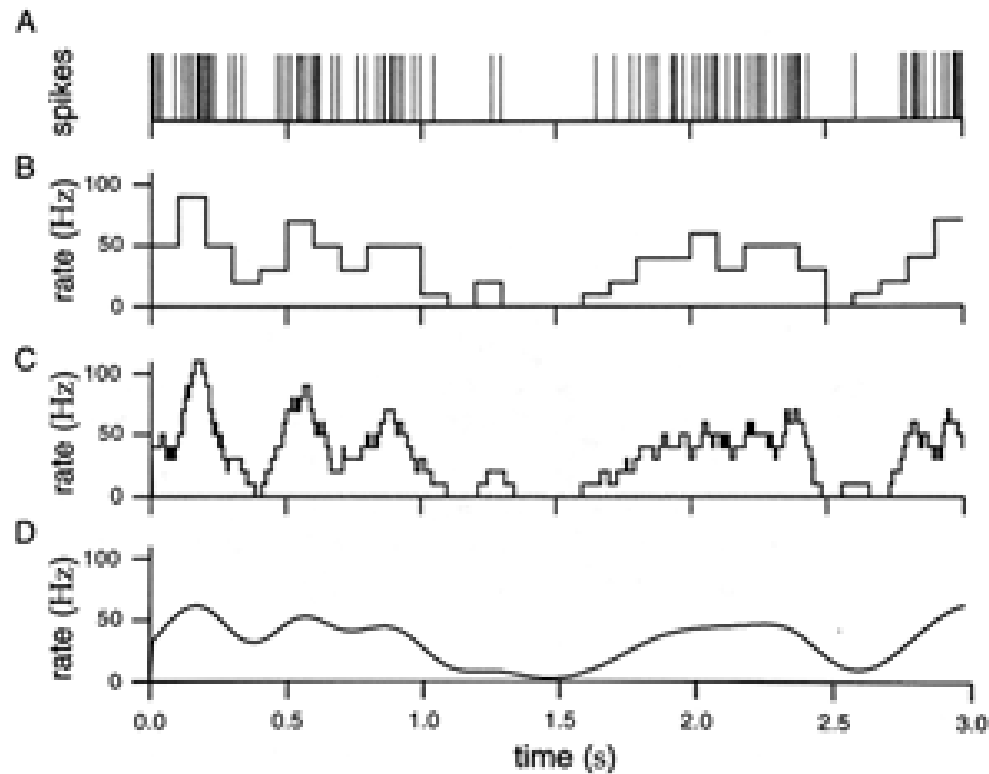
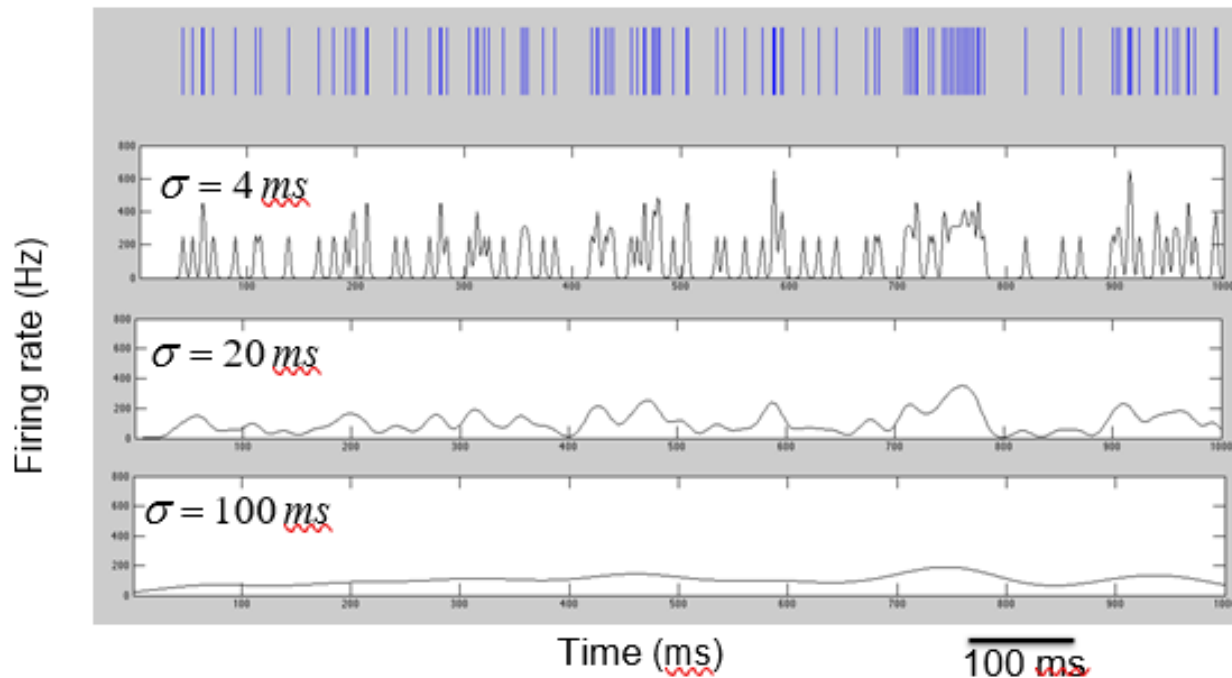


Figure courtesy MIT Press. From Dayan, P., and L. Abbott. [*Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems*](#). 2001.

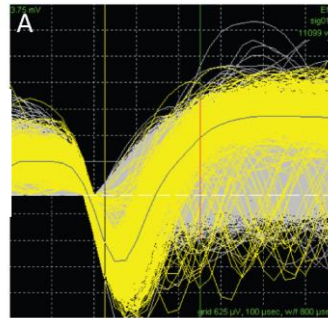
Quantifying firing rates

You have to choose a timescale to measure the firing rate. But you get a different answer for every different timescale

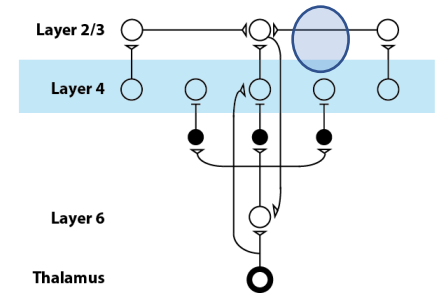


Towards a cortical map? / Single unit recording

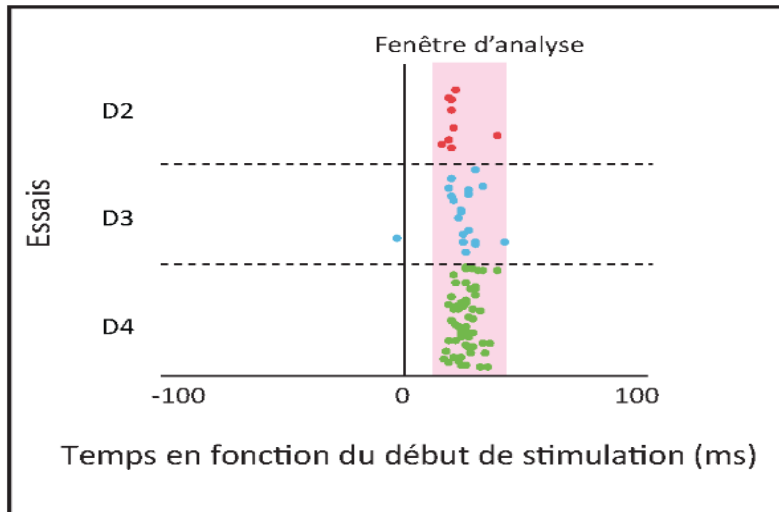
Electrophysiology : Single unit recording in Layer 4



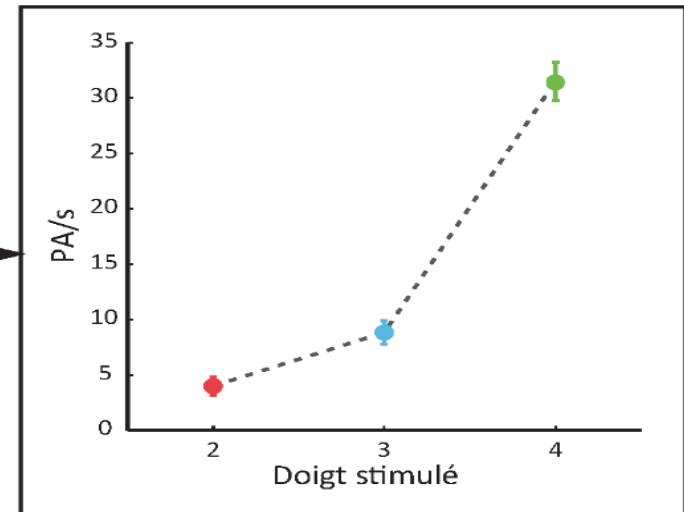
Alignement temporel
sur les stimulations
pour chaque essai



D

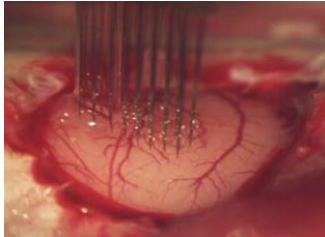


E

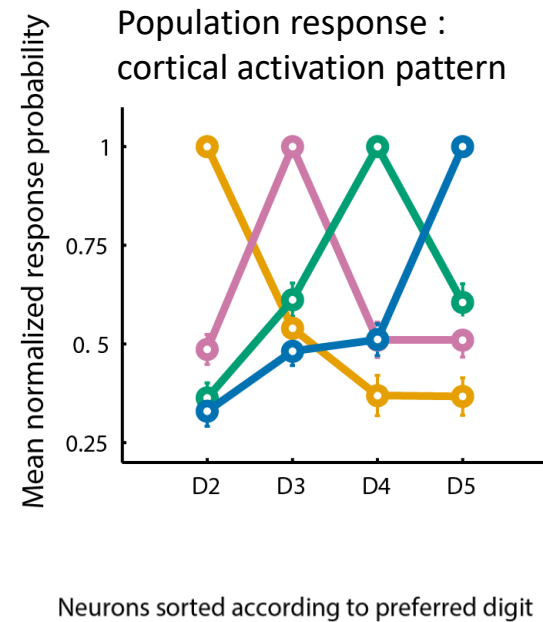
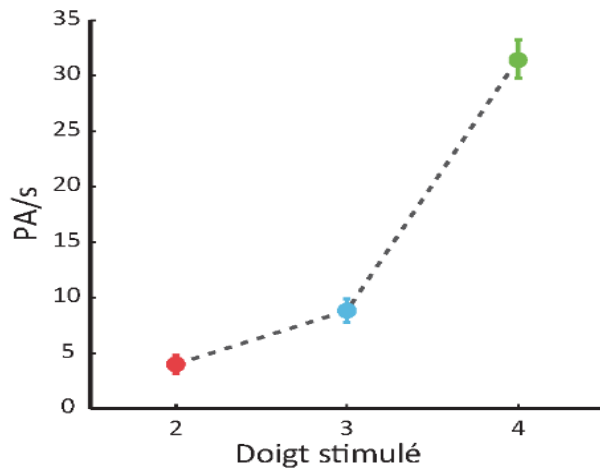
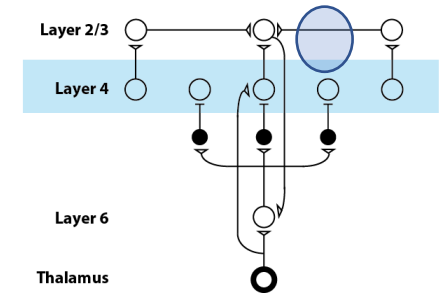


Courbe d'accord

Towards a cortical map? / Single unit recording

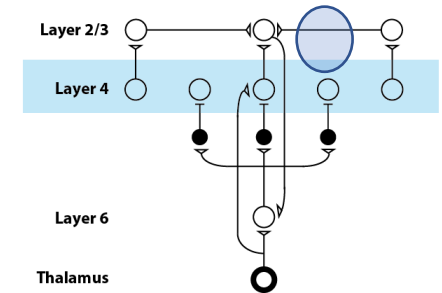
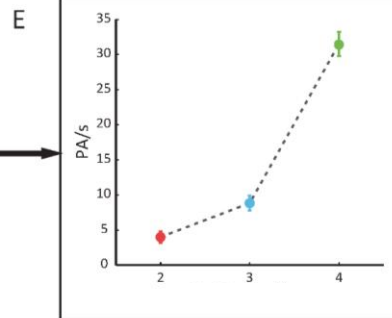
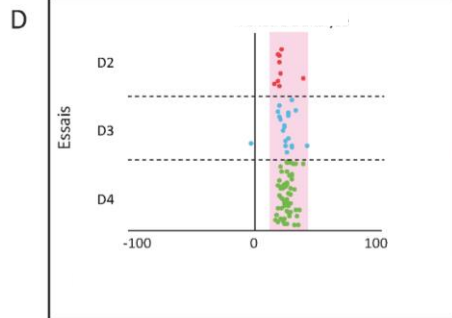
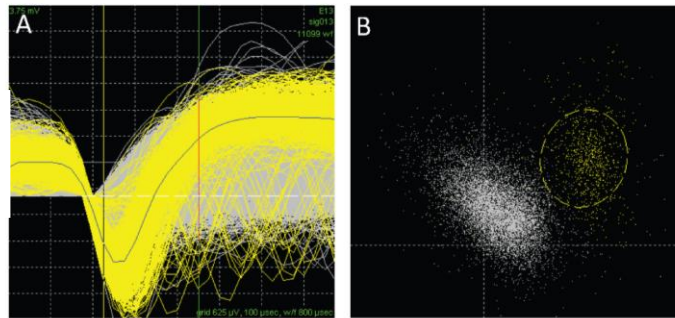
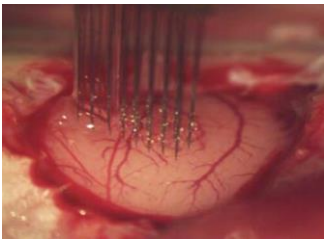


Electrophysiology : Single unit recording in Layer 4

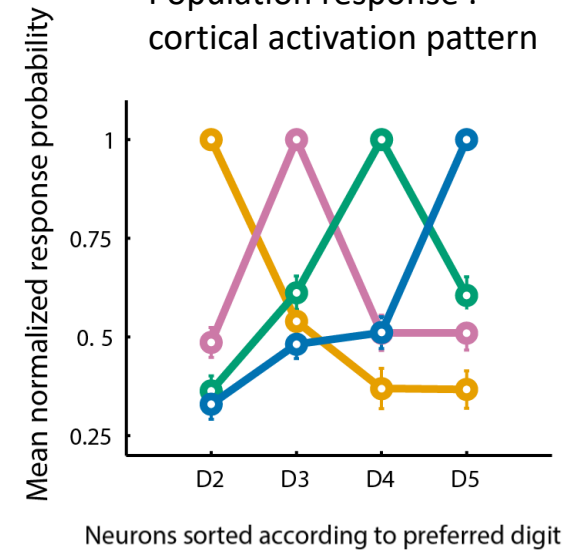


Towards a cortical map? / Single unit recording

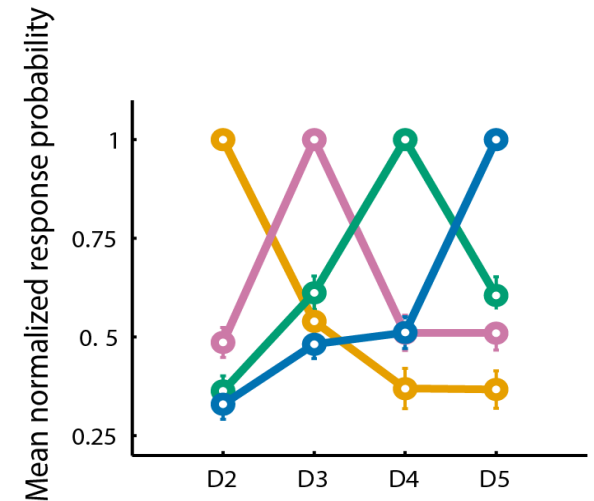
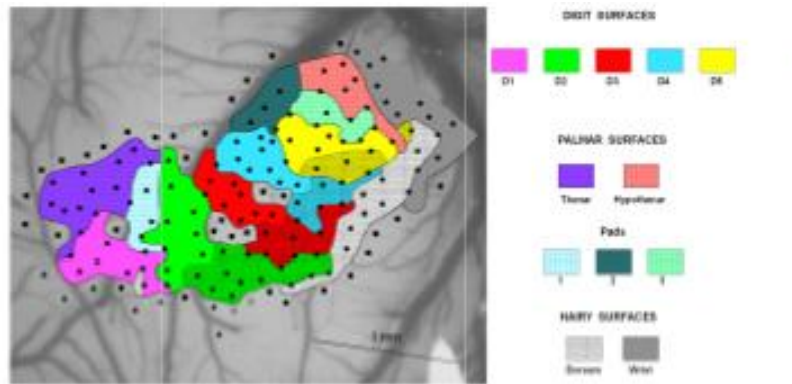
Electrophysiology : Single unit recording in Layer 4



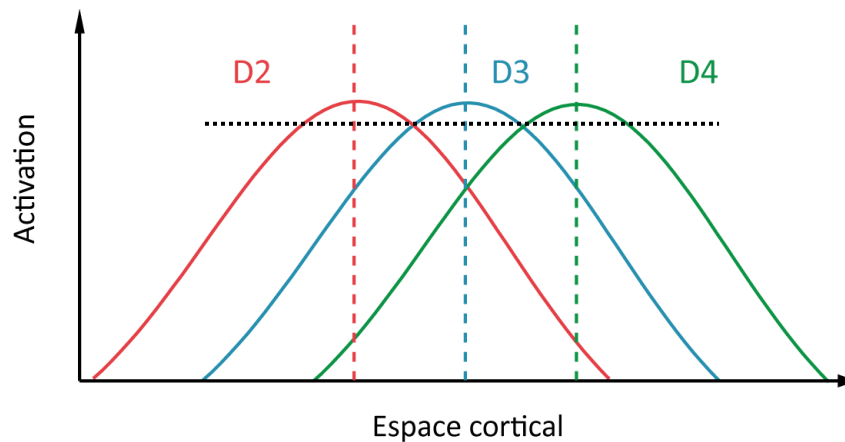
Population response :
cortical activation pattern



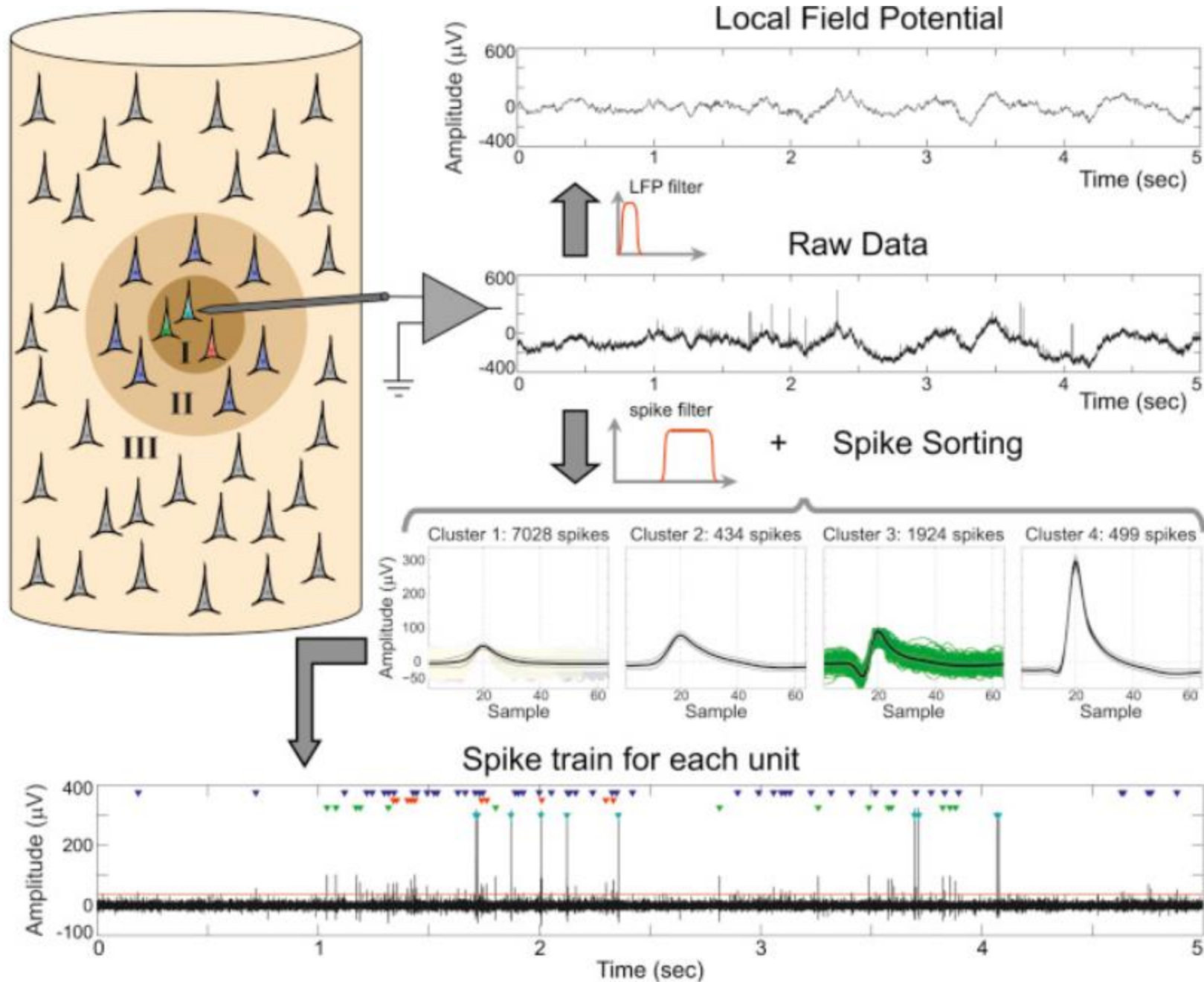
Towards a cortical map? / Single unit recording



Neurons sorted according to preferred digit



Corbo et al. (eNeuro, 2018)

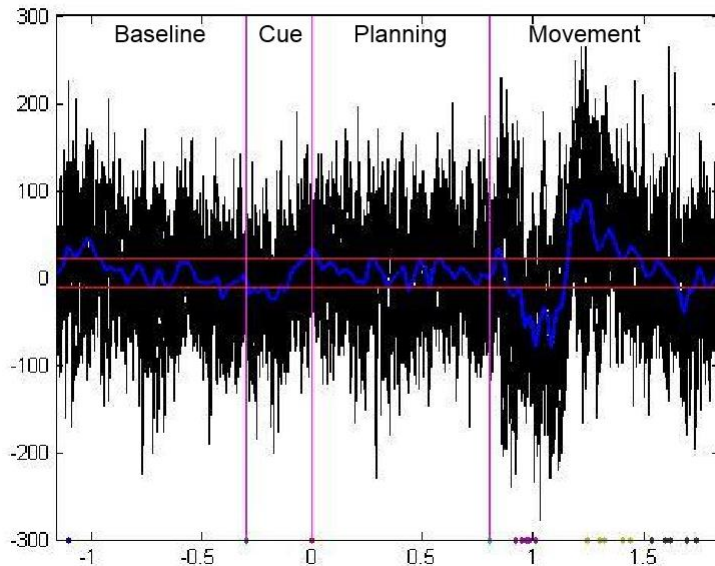


Towards a cortical map? / LFP

What is the local field potential

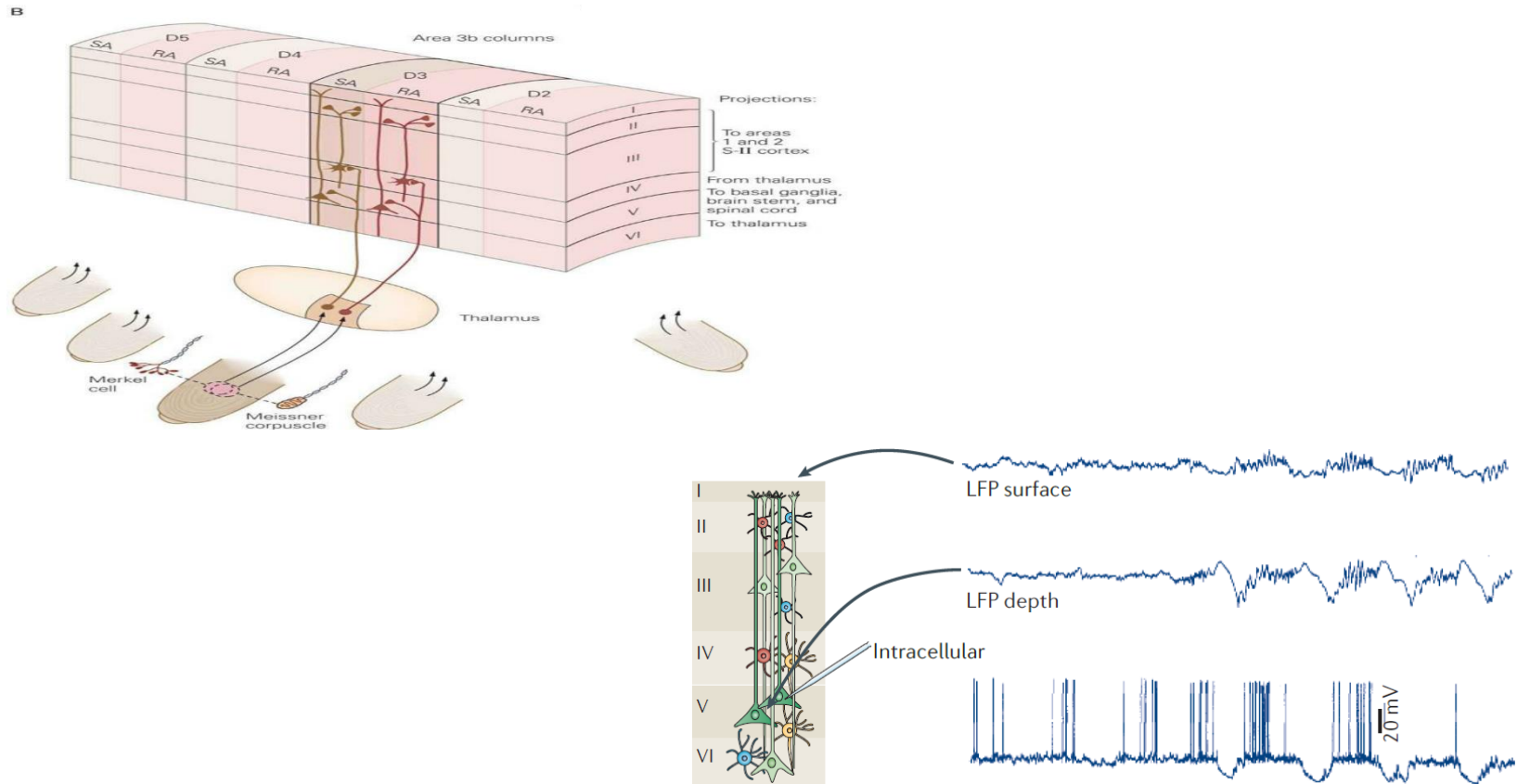
- “The local field potential is the extracellular current flow that reflects [...] the linearly summed postsynaptic potentials from local cell groups”(Buzsaki, 2004).
- has temporal structure mainly in the frequency range of 0-100 Hz
- is correlated with the fMRI(BOLD) signal

LFP evoked potential



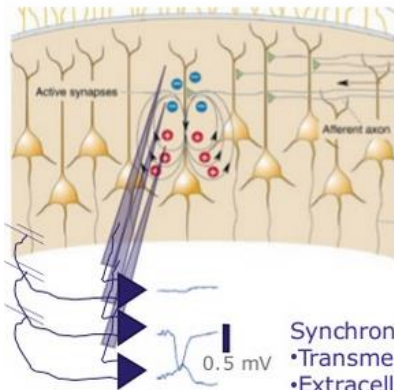
Towards a cortical map? / LFP

Electrophysiology : Local Field Potentials in Layer 4

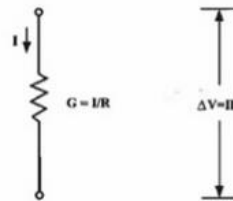


Towards a cortical map? / LFP

What generates the evoked field potential ?

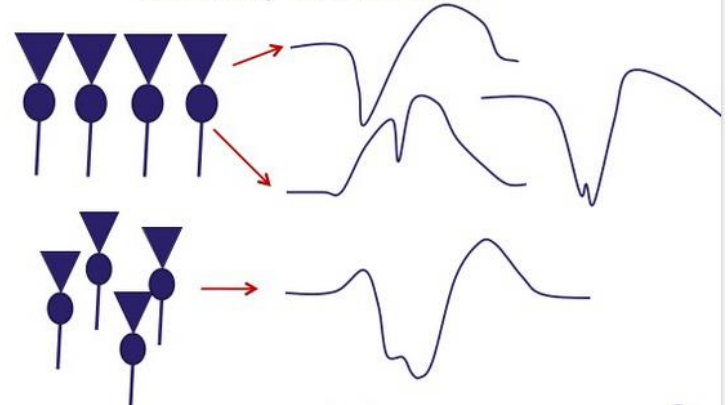


$$\Delta V = IR = I/G \text{ (units: volts)} \quad (2)$$



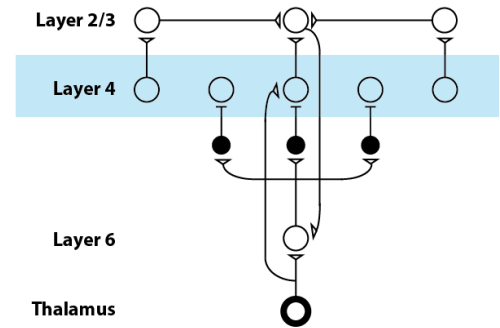
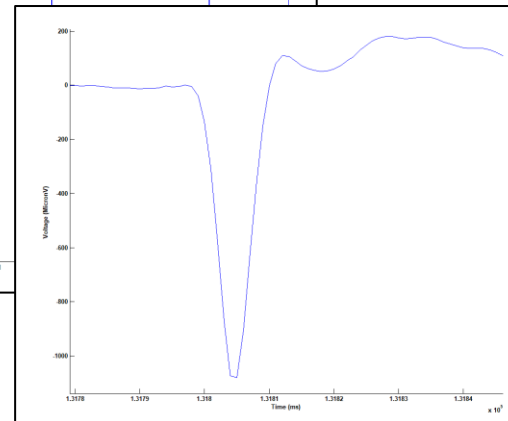
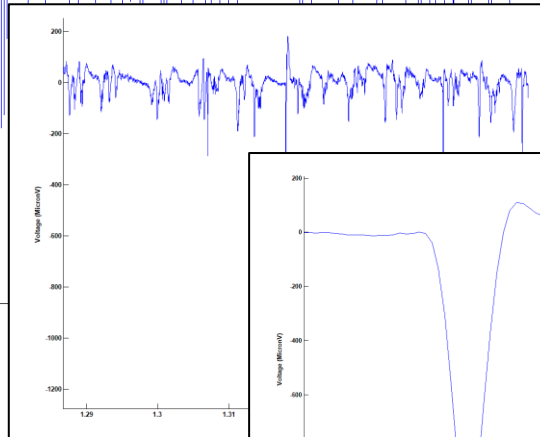
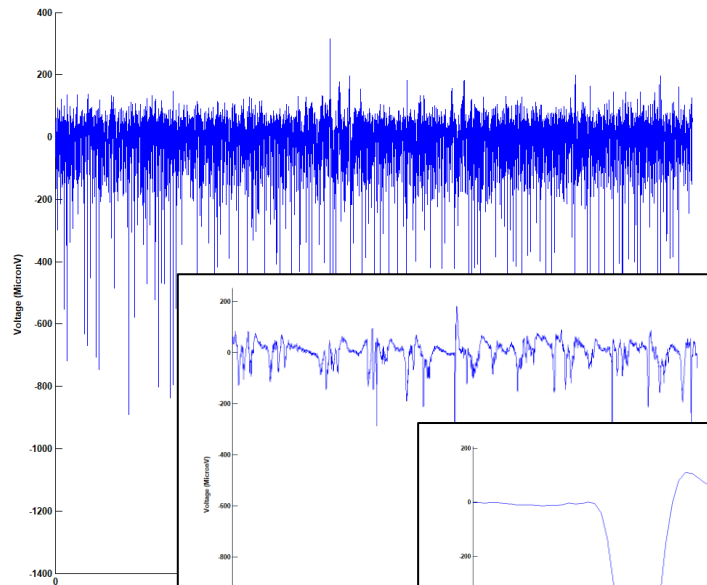
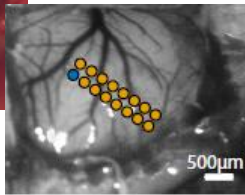
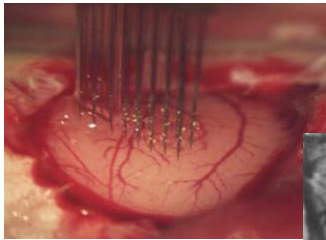
Synchronic activation:
• Transmembrane current flow
• Extracellular current flow and the resistant properties of the extracellular media $\rightarrow$ voltage changes in the field potential

Shape of evoked field potentials as function of anatomy and location



Towards a cortical map? / LFP

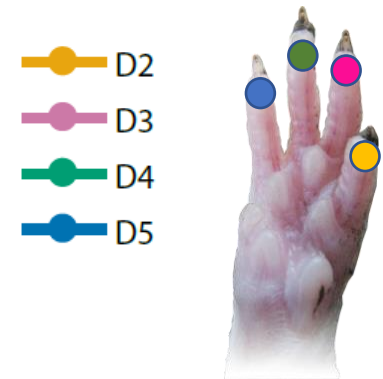
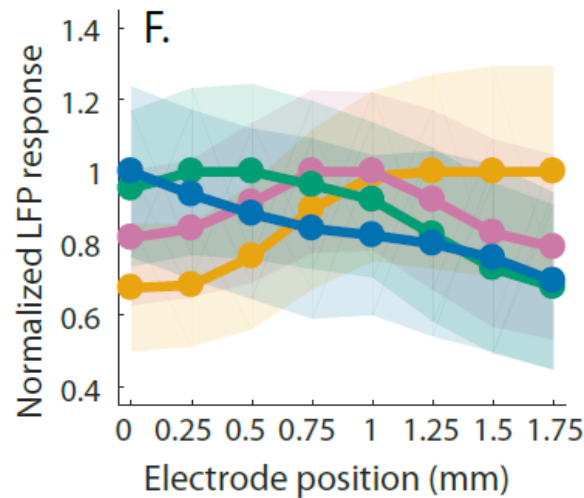
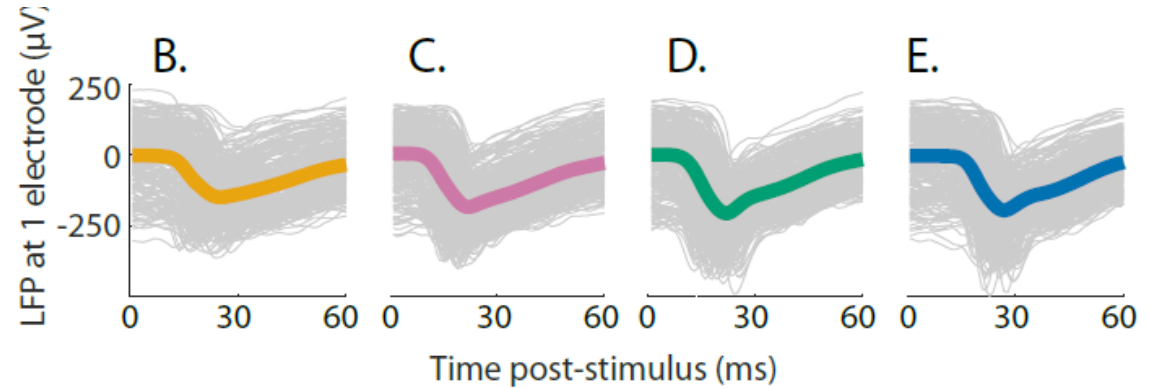
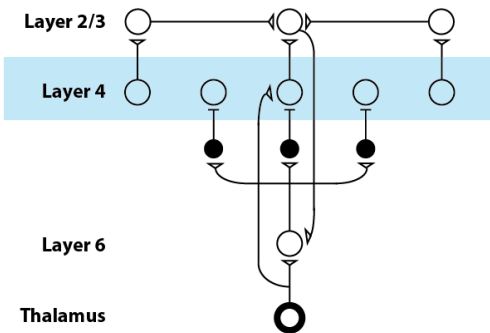
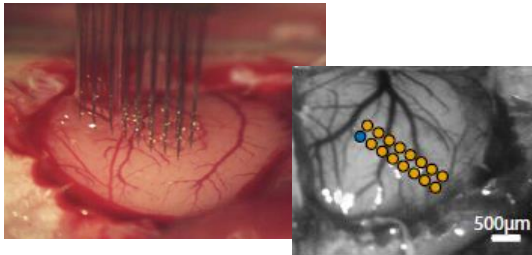
Electrophysiology : Local Field Potentials in Layer 4



Adapted from Raizada & Grossberg, 2003

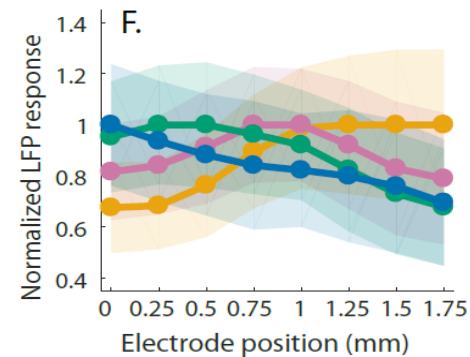
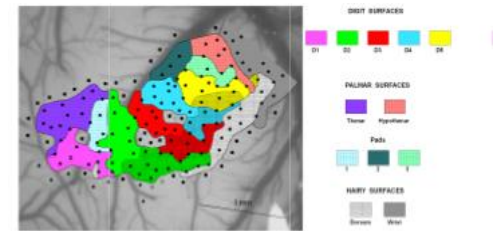
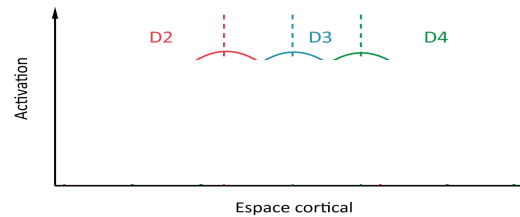
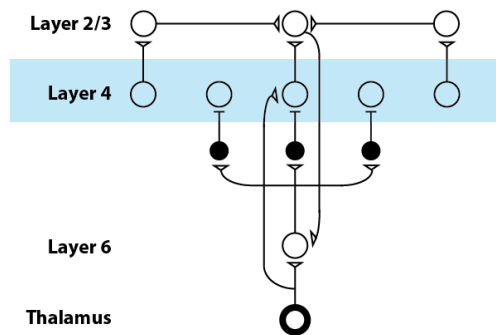
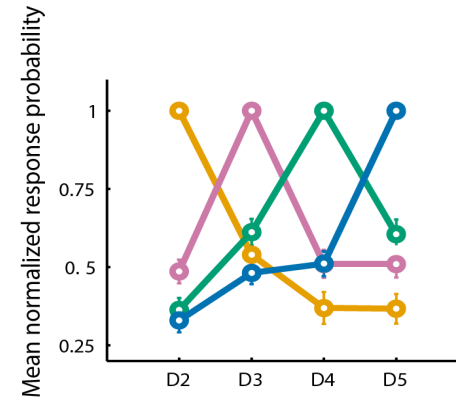
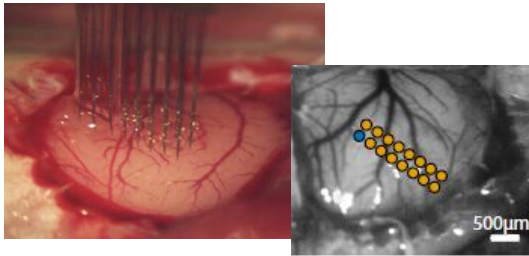
Towards a cortical map? / LFP

Electrophysiology : Local Field Potentials in Layer 4



Towards a cortical map? / LFP

Electrophysiology : MUA, Single Unit and Local Field Potentials in Layer 4

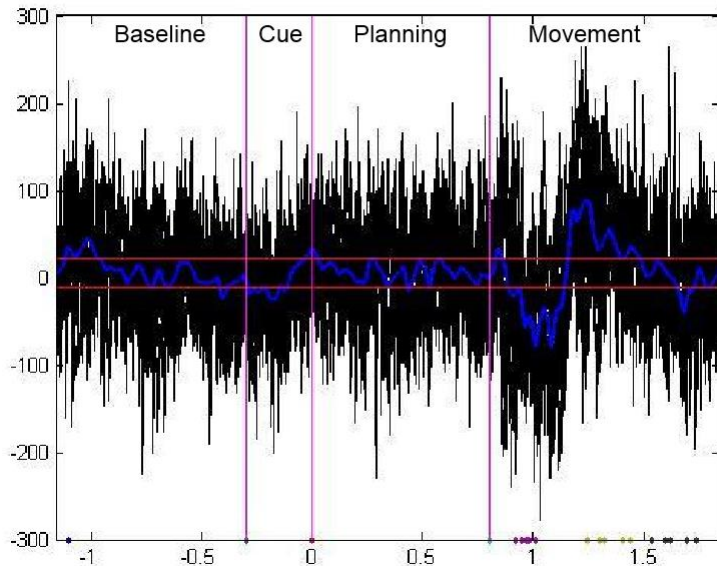


Towards a cortical map? / LFP

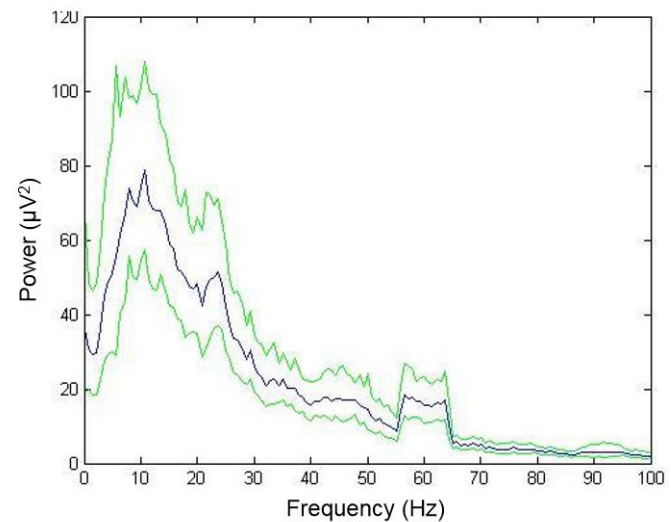
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LFP evoked potential

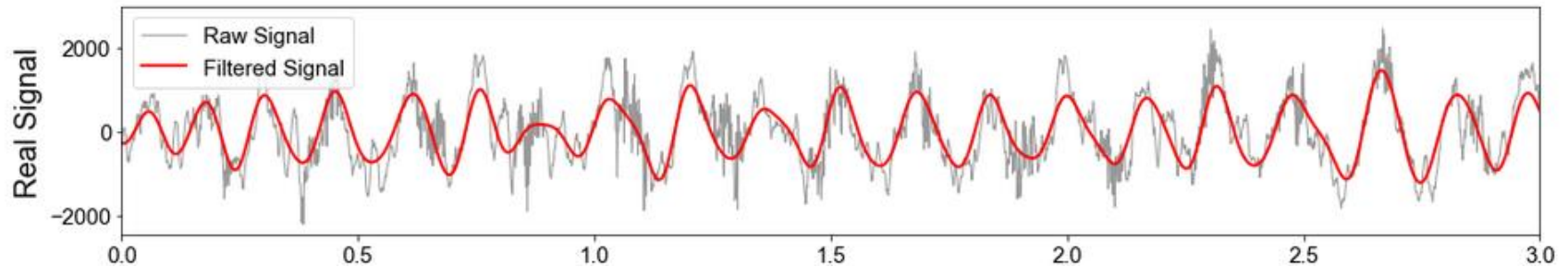


LFP spectrum

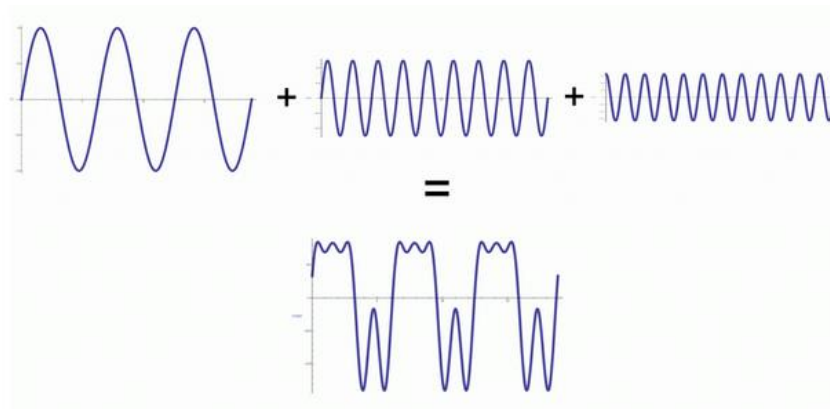


Towards a cortical map? / LFP

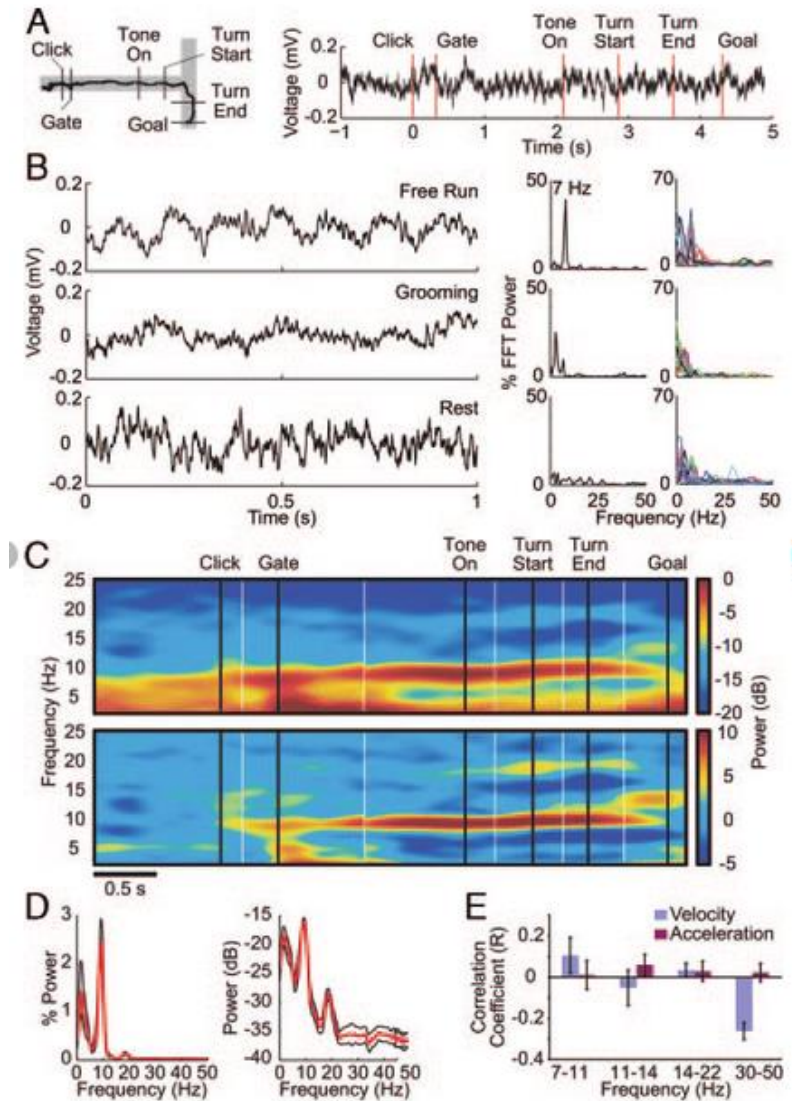
Oscillation : from Raw signal to frequency bands



Notion de transformée de Fourier



Towards a cortical map? / LFP



Towards a cortical map? / LFP

Behavioral/Systems/Cognitive

In Primary auditory cortex

Robustness of Cortical Topography across Fields, Laminae, Anesthetic States, and Neurophysiological Signal Types

Gao et al. • Auditory Cortex Topography

J. Neurosci., July 4, 2012 • 32(27):9159–9172 • 9167

